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Showing Off or Laying Low? The Economics of Psych-outs†

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We analyze the incentives for showing off, which we model as a costly signaling game, and study the consequences of norms against such behavior. Prior to competing in a contest, a newcomer can signal his talent to an incumbent. In equilibrium, costly signaling of ability occurs only when the newcomer is exceptionally talented. In such situations signaling benefits both contestants: the newcomer for obvious reasons; the incumbent by economizing on wasted effort in the contest. Our results rationalize the emergence of norms against showing off in settings where total effort is important. When selection efficiency matters, such norms decrease welfare. (*JEL* D82, D83, D91, Z13)

We study situations in which an incumbent faces a newcomer in a competition. Each competitor decides how hard to compete and, in the end, the better performer is awarded a prize. Whether in the workplace, the political arena, high school, or college, such situations abound.[1] Oftentimes, before the start of the competition, the newcomer has an opportunity to signal—or show off—his talent. A good use of this strategy will soften subsequent competition, while a wrong

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[1] For example, many workplaces feature some form of competitive incentives. For instance, Cowgill (2015) shows that relative performance evaluation co-determines wages for 77 percent of workers in the United States. Such competition motivates effort, but the nature of incentives depends on relative ability as well as beliefs about a competitor's relative ability (Brown 2011). Horizontal movement of workers is commonplace. For example, Fallick et al. (2006) document that between 1994 and 2001 more than 2.4 percent of employees in Silicon Valley changed their employer within the valley on a monthly basis. Consequently, asymmetries in the information about a coworker's talent are important: information about a new hire's talent is inherently less precise.

decision may lead to an escalation of rivalry. In this paper, we investigate the economics of showing off, laying low, and psych-outs.

In our archetypal situation, a newcomer competes against an incumbent for a prize. Both know the incumbent's ability, but the newcomer's ability is unknown to the incumbent. Prior to the competition, the newcomer can try to reveal his talent through a costly signal. Signaling is cheaper, the more talented the newcomer. We interpret sending such a costly signal as showing off. We find that newcomers are often better served by not investing in signaling and laying low. Only newcomers that are exceptional, relative to the incumbent, are able to benefit from revealing their type in equilibrium.

The point of showing off is to convince the incumbent to compete less intensely. Whether revealing talent achieves that depends on relative ability. Newcomers that are similar in ability to the incumbent motivate the incumbent to compete harder. These newcomers are well advised to lay low, i.e., to send no costly type revealing signal. Very low ability newcomers appease the incumbent and thus have an incentive to reveal their talent. But for them signaling is very costly, and they are not able to credibly reveal their type in equilibrium. It is the exceptional newcomers who discourage—or psych out—competition by showing off and for whom showing off is relatively cheap at the same time. We identify the talent threshold where psych outs become possible and study its properties.

Finally, we examine the economics of showing off from a societal perspective. We show that when society cares about aggregate effort expended in the contest, showing off should be strongly discouraged. In many cultures this discouragement takes the form of a social norm against showing off. For example, the English-speaking world uses the metaphor of "Cutting Down the Tall Poppies."[2] In schools, "tall poppies" are straight-A students who are voluble in demonstrating their intelligence. In the workplace, they are colleagues who showcase their expertise during water cooler discussions and lunch breaks. Cutting down the tall poppies means punishing and thus discouraging showing off behavior. Our analysis reveals that such norms can be functional and create social value. By limiting showing off opportunities, society shuts down this channel of discouraging others as a means to economize on effort.

Casual empiricism, though, suggests exceptions to this rule: Silicon Valley start-ups seem to owe much of their success to "letting the tall poppies bloom." In these settings, talent identification and retention determines success. We show that a norm against showing off is detrimental when selecting talent is the goal. Intuitively, showing off by a talented newcomer psychs out the incumbent and thus lowers her effort which benefits selection efficiency. Thus, the same model of showing off can account for apparently vastly different behavioral norms.

To be precise, we amend the standard contest model to allow for pre-contest costly signaling of ability in a situation where information is asymmetric—one party's abilities are known to all while the other's are unknown and private knowledge.

[2] This metaphor dates back to accounts by Herodotus that describe how Thrasybulus led a herald from Periander through a wheat field. In response to the herald's question of how he would most safely govern his city, Thrasybulus silently cuts down the tallest ears of wheat. (Herodotus, The Histories, Book 5, 92-f).

Signaling costs are perfectly correlated with ability. In this setting, our main results are the following:

- There always exists a threshold equilibrium in which, if any, only newcomer types that are exceptionally talented signal and reveal their type. Less talented newcomers' types remain unknown.
- Showing off and laying low are used by competitors to soften competition to their mutual benefit. Doing so, however, harms a society mainly interested in total effort. A norm against showing off may serve as a remedy and increases aggregate effort.
- The possibility of showing off tends to accentuate competitors' differences and thereby improves selection efficiency. Thus, a norm against showing off is detrimental to a society interested in selecting the most able competitor.

We now place our findings in the context of the extant literature. Classical signaling games in the tradition of Spence (1973) also deal with the question of whether to show off (see e.g., Feltovich et al. 2002, or Daley and Green 2014, for some more recent contributions to this strand of literature). In contrast to our paper, here the signaling is addressed to a principal and not to a competitor. Another classic literature related in spirit considers the strategic investment decision of an incumbent firm in anticipation of potential market entry (e.g., Fudenberg and Tirole 1984) and classifies optimal strategies by strategic complementarity or substitutability of the players' choices. In our setting with incomplete information actions can be at the same time strategic complements against one type and strategic substitutes against another, as best-response functions are not monotone. This is a standard feature of many contest models which are a natural setting to study signaling but does not fit immediately into the categorization of this literature.

The extant contest literature mainly focuses on driving effort or selection by manipulating various aspects of the contest such as prize amounts, numbers of players, and information disclosed by the contest organizer. We focus on the equilibrium "culture" of information revelation that prevails among competitors.

More concretely, a large branch of the literature studies contest structure including the allocation of prize money and timing in the resolution mechanics of the contest itself.[3] More closely related, the paper relates to the literature studying asymmetric contests, for example Nti (1999); Stein (2002); Cornes and Hartley (2005); Siegel (2010); or Franke et al. (2018), which establishes that a contestant is most aggressive when evenly matched, which also drives signaling incentives in our setting. The effect of uncertainty about a rival's strength is analyzed for example in Hurley and Shogren (1998) who find that the uninformed rival experiences a decline in his chance of success relative to full information. Wärneryd (2003) studies asymmetric information regarding the commonly shared value contestants attach

[3] For example, Moldovanu and Sela (2001), Clark and Riis (1998), or Sisak (2009) focus on optimal prize division with a given budget. Gradstein and Konrad (1999) and Moldovanu and Sela (2006) analyze one-shot contests versus tournaments with multiple stages. Yildirim (2005), Klumpp and Polborn (2006), Konrad and Kovenock (2009), Denter and Sisak (2015, 2016), or Seel and Strack (2016) focus on dynamics in effort choices where the same competitors expend effort repeatedly against each other.

to winning the prize, and how this impacts contestants' aggregate effort. Fu (2006) analyzes how informational asymmetries change the endogenous order of moves in contests and how this impacts efforts. Building onto this literature another branch studies how the principal might wish to manipulate the information competitors have about one another; for example, by releasing information about the competitors' abilities before the start of the competition. Leading examples of this branch are Ederer (2010), Klein and Schmutzler (2017), Zhang and Zhou (2016), Lu et al. (2016), or Crutzen et al. (2013). Relatedly, Olszewski and Siegel (2019) focus on the decision of a principal to disclose information about (college admission) contest outcomes and how this affects student effort choices and welfare. We build on the results in Hurley and Shogren (1998) to study the incentives of the informed rival to signal his talent as well as the broader implications of this choice. Thus we focus on the choices that would be made when the agent, rather than the principal, decides.

There is a small literature that also focuses on disclosure by agents in different settings.[4] Wu and Zheng (2017), Kovenock et al. (2015), as well as Denter et al. (2014) study ex ante commitment by the competitors to disclose information costlessly. Kovenock et al. (2015), who study an all-pay auction setting, also consider an extension where information sharing choices are made interim, i.e., after competitors received their private information. Ewerhart and Grünseis (2018) consider the analogous case of a probabilistic contest. In contrast to our setting, disclosure is always costless and truthful in these papers. Costly signaling is considered in Katsenos (2009) and Ifergane and Sela (2017), both using a binary type Tullock contest model. Katsenos (2009) studies agents' incentives to signal their types when there is two-sided asymmetric information, while Ifergane and Sela (2017) study one-sided asymmetric information, but, different from our study, assume that signaling costs are not correlated with type, comparing the Tullock contest to an all-pay auction setting. Finally, another small literature considers the case where pre-contest communication affects contest payoffs directly and not just indirectly through information transmission. Fu et al. (2013) and Yildirim (2018) study pre-contest communication, i.e., a statement of confidence, that increases the cost of losing the contest. Heijnen and Schoonbeek (2017) study the case where the informed player can commit to reduce the prize either by some absolute amount of money or proportionally, conditional on winning it. In contrast, pre-contest communication in our model affects contest payoffs only through the strategic effect on the rival's effort. Finally, Hörner and Sahuguet (2007) study a competitor's incentives to manipulate information in an auction setting through an aggressive or cautious opening bid. We add to this literature by studying costly signaling in a Tullock model allowing for arbitrary type distributions and considering a setting where signaling costs are correlated with types. The main purpose of our study, different to the extant contest literature, is to study how norms regarding signaling—or showing off—affect expected selection efficiency and expected aggregate efforts. To the best of our knowledge we are the

[4] Information may of course not only be exchanged through disclosure, but an uninformed agent may as well engage in information acquisition instead. Morath and Münster (2013) study the incentives to *acquire* relevant information about a contender prior to a contest.

first to rationalize the emergence of norms against showing off and put forth conditions that make the emergence of such norms favorable.

While the question whether to show off is clearly relevant for human interactions, animals also show off to signal their talent. For instance, scientists have studied the phenomenon of roaring behavior by stags during the red deer's mating season. A challenger initiates roaring against the incumbent harem holder. After assessing one another during this roaring strength display, the red deer stags decide whether to enter into direct conflict (see for example Payne and Pagel 1997). Critically, the incumbent's subsequent aggressiveness depends on the information gained in the roaring phase of the meeting. Recent studies found that also octopi engage in costly signaling before potentially entering into conflict. They find that octopus colouring and body posture are correlated with escalation of disputes (Scheel et al. 2016). The testable predictions we offer apply as much to animal as human behavior.

There is a vast literature examining how institutions and norms form, and how they impact policy.[5] Many scholars argue that norms correct for positive or negative externalities and thus a norm is likely to arise in settings where it is socially beneficial (see e.g., Fehr and Fischbacher 2004). Relatedly, a behavioral regularity noted in the experimental economics literature is that a more socially beneficial norm of behavior (in a broad sense) is more likely to be adopted.[6]

We are agnostic on how norms emerge. Rather, we take as given that the likelihood of a norm prevailing depends on its social desirability. Our contribution, relative to the extant literature, is to apply this metric to showing off situations embedded in contests. As described above, norms against showing off are prevalent in many contexts, though they are noticeably absent in others. To the best of our knowledge we are the first to offer a rationalization for the emergence of norms against showing off in certain settings but not others. Applying this notion of norm emergence, we find that a norm against showing off is more likely to emerge when society benefits from aggregate effort expended in the contest and less likely when selection of talent is society's primary goal.

The paper is organized as follows. The next section, Section I, introduces the model. Section II features the main analysis where we study showing off in the form of costly signaling by the newcomer in Subsection IIA. In Subsection IIB we study how the incumbent is affected by costly signaling. In Section III we study the effects of a norm against showing off which we model as a fixed cost of signaling. Finally, in Section IV we study the implications of "productive" signaling, i.e., signaling that not only reveals information but also translates into effective effort. Section V concludes. All proofs are relegated to the Appendix.

[5] See, e.g., Acemoglu and Robinson (2016) for the importance of norms for the emergence of different institutions, Sliwka (2007) for how norms affect organizational design, Name Correa and Yildirim (2016) for how norms affect charitable giving, or Bowles and Polania-Reyes (2012) for a survey.

[6] Zelmer (2003) provides a meta analysis of giving behavior in linear public goods games where a higher marginal per capita return tends to increase contributions, while Parravano and Poulsen (2015) show that stake size increases coordination on label-salient strategies for symmetric coordination games.

I. The Model

As argued above, asymmetries in information for example about a co-worker's talent are important, since information about a new hire's talent is inherently less precise than information about well established, or incumbent, competitors. The model we construct is a parsimonious representation of such an environment that serves to highlight the main forces at play. Two competitors vie for a prize, which goes to the better performer and whose value we normalize at 1. The losing competitor receives a consolation prize, which we normalize at zero.[7] These competitors could be two middle managers competing for a bonus, political rivals competing for office, or students competing to be valedictorian.

Let x_i denote competitor i's effort, where both players choose effort simultaneously. When two managers compete, this effort can be thought of as the thoroughness of the market research each of them conducts to use as the basis for decisions that affect the success of the company. A competitor's realized "performance" is a combination of noise and effort; thus, it may well be the case that the competitor who exerts more effort still loses the contest. Formally, let ϵ_i be the noise component of performance and suppose that measured performance takes the form:

$$y_i = x_i \cdot \epsilon_i.$$

We assume that ϵ_i is drawn from a Weibull (maximum) distribution with a mean equal to one. Thus, on average, a player's performance corresponds to her effort.

The principal awards the prize to the better performing competitor, i.e., the competitor with the higher value of y_i. Thus, while effort is helpful in securing the prize, exerting more effort than a rival does not guarantee success. Our formulation of the stochastic component of performance produces a convenient specification relating a competitor's chance of success to her effort. In particular, the chances of success are[8]

$$p_i = \frac{x_i}{x_i + x_j} \quad \text{and} \quad p_j = 1 - p_i. \tag{1}$$

This formulation, often called the lottery contest success function, proves extremely tractable, so much so that it is virtually standard in examining contests and tournaments.[9]

Competitors are risk neutral and differ in ability/talent. We model this difference by assuming that a more talented person has a lower marginal cost of effort, c_i. Returning to our leading example, a more talented manager can produce reports

[7] One may worry about normalizing both the winning and losing prize, but, in a contest setting, this is without loss of generality. Effort in a contest depends solely on the spread between the value of the prizes, rather than their level; thus, we have essentially normalized the spread to be equal to one "util." Normalizing the prize spread is also without loss of generality as we allow for general constant marginal costs of effort. A linear transformation of the utility function, i.e., dividing by a constant, leaves strategic incentives intact.

[8] See Fu and Lu (2012) for details on how to derive this specification.

[9] The functional form of equation (1) is problematic when both players choose zero effort. Following the literature, we assume that a coin flip determines the winner in this case. See, for example, Konrad (2009) or Sisak (2009) for surveys of the vast literature using this specification.

with a given level of thoroughness more quickly than a less talented manager, i.e., she incurs lower costs.

While each competitor knows her own marginal costs (ability) at the time of effort choice, information about the other player may differ. In particular, we assume that the talent of competitor I, the incumbent, is common knowledge, whereas the talent of her rival, the newcomer, whose identity we denote by N, is his private information. The asymmetry describes common hiring situations where an internal candidate faces off against an external candidate or where a new salesperson competes against a veteran incumbent. It also serves the additional purpose of highlighting the main forces in the economics of "showing off," which are only relevant in private information settings, while minimizing the complexity of the analysis. Note that some settings are better described by assuming that it is the incumbent who possesses private information regarding her type, for example when learning on the job is the most important determinant of performance and this is not observable from the outside. Our analysis and results also apply to these situations after adequate relabeling of players.[10]

Prior to both parties' contest effort choices, the newcomer has the opportunity to choose any signaling effort level $s_N \geq 0$ at cost $c_N s_N$, i.e., to engage in costly but non-performance related displays meant to signal his ability.[11] Such displays can take many forms. In a corporate setting, they might consist of casual conversations with co-workers revealing deep industry knowledge or especially impressive presentations at group meetings. Importantly, the cost of such displays varies with the newcomer's talent. While a talented individual can draw on his skills, a less talented individual might have to spend the whole night preparing the anecdote he plans to present effortlessly the next day. For simplicity, we assume that the marginal cost of signaling effort is equal to the marginal cost of contest effort, c_N.

Since the incumbent does not know the newcomer's talent, signaling effort choices of the newcomer can affect the incumbent's beliefs and hence alter the strategic circumstances of contest effort choices. Studying when different newcomer types choose different signaling efforts, and thus newcomer types at least partially separate and some information about the newcomer's talent is revealed, and how this impacts subsequent contest effort choices, forms the heart of the paper.[12] Figure 1 summarizes the timing of moves.

[10] Our results readily extend to situations of two-sided asymmetric information where ability distributions differ. In particular, our results may be thought of as describing the limiting value of a sequence of equilibria where the ability of one player converges to the Dirac distribution. Katsenos (2009), on the other hand, offers some insights on the limiting case where ability distributions are the same, i.e., either competitor is of type c_L with probability q and type c_H with probability $1 - q$. We discuss his findings and relate them to our results in Subsection IIA.

[11] In Section IV we relax this assumption and allow for directly productive signaling effort.

[12] Note that we study a *costly* signaling game. Alternatively, signaling may be costless, in which case we would have a cheap talk model (Crawford and Sobel 1982). Under this alternative assumption, no (effort-relevant) information revelation is possible in our framework. The reason is that for each newcomer type, lower effort of the incumbent is strictly preferred. Thus, as we discuss below, unless when $c_L = c_I^2/c_H$, there always exists a unique newcomer type against which the incumbent would choose minimal effort under complete information. All newcomer types always want to convince the incumbent to be of this type. Thus, cheap talk cannot reveal information about the newcomer's type. For this reason, we focus on situations where signaling is costly for the newcomer and this cost is correlated with his type. Many settings have this flavor, such as a coworker signaling his knowledge about the industry, a scholar discussing the work of others, or a politician holding an inspiring speech.

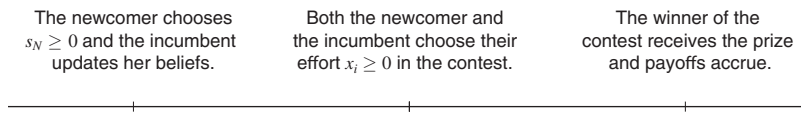

FIGURE 1. SEQUENCE OF MOVES

Formally, the incumbent I's cost parameter c_I is common knowledge at the beginning of the game, while the newcomer's cost c_N is a random variable from the perspective of the incumbent with a commonly known CDF $F(c_N)$ over some interval $\mathcal{S}_N = [c_L, c_H]$. We assume that $F(c_N)$ is twice differentiable and strictly increasing on $\mathcal{S}_N = [c_L, c_H]$. Though we state most of our results for continuous CDFs, we derive further results as well as illustrate the intuition and economic forces using a two-type model, i.e., when $\mathcal{S}_N = \mathcal{S}_N^B = \{c_L, c_H\}$ and $f(c_N) = \mathbf{q}^2 = (q, 1-q)$, with $q \in (0,1)$.[13]

Throughout we focus on interior equilibria, that is equilibria in which all types of both players choose positive effort in the contest. A sufficient condition for interiority is the following.[14]

ASSUMPTION 1. $2c_L \geq c_H \geq c_I \geq c_L$.

As usual, there may be multiple Perfect Bayesian Equilibria (PBE) in signaling games. Since we study a rich type space with more than two possible newcomer types, refinements such as the intuitive criterion are not sufficient to make unique equilibrium predictions.[15] To be able to effectively narrow down the set of equilibria, we restrict off equilibrium beliefs by using another standard refinement, namely the D1 criterion due to Cho and Kreps (1987).[16] When referring to "equilibrium" in the following we thus imply a PBE that satisfies D1. We discuss how our main results are affected by this assumption after Proposition 2 and Theorem 3.

[13] Note that this binary scenario does not satisfy the formal conditions on $F(c_N)$.

[14] Corner solutions are most likely to arise when the probability mass is centered around c_L, and c_I is relatively close to c_L. In such cases, weaker newcomer types may optimally choose zero effort in case the incumbent does not learn the newcomer's type. Assumption 1 rules out corner solutions by restricting the maximum difference between highest and lowest newcomer type. Note that most of our results also apply under a less strict version of this assumption, $4c_L \geq c_H \geq c_I \geq c_L$; only Proposition 4 and Theorem 3 rely on it. For details please consult Appendices H and J. Also note that in our baseline setting the incumbent will never be discouraged from investing effort in the subsequent competition regardless of beliefs induced through the costly signal. This is possible when signaling effort also is productive in the contest, which is discussed in Section IV.

[15] See, for example, Chapter 13 of Mas-Colell, Whinston, and Green (1995). With more than two types, the Intuitive Criterion by Cho and Kreps (1987) fails to make unique predictions, and other refinements, such as D1 by Cho and Kreps (1987) or the related Universal Divinity by Banks and Sobel (1987), are needed to narrow down the set of reasonable equilibria.

[16] D1 is a widely used equilibrium refinement; see, for example Eső and Schummer (2004), Kamenica (2008), Andreoni and Bernheim (2009), Adriani and Deidda (2009), Daughety and Reinganum (2014), Belenzon et al. (2017), or Hedlund (2017). It restricts off equilibrium beliefs by requiring the receiver to assign probability 1 to the type who has an incentive to deviate in the greatest number of circumstances. It is thus slightly weaker than Universal Divinity by Banks and Sobel (1987), see also the discussion in Section 5 of Kreps and Sobel (1994).

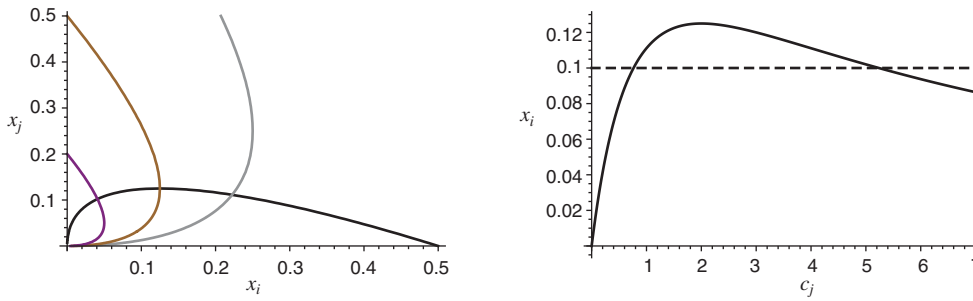

FIGURE 2

Notes: A player's effort is maximal when she faces a competitor of identical talent and decreases strictly when the match gets more unbalanced. Left panel: Best response functions for i and j, given $c_j = 2$ and $c_i \in \{1,2,5\}$. Right panel: Equilibrium effort of i with $c_i = 2$ as a function of $c_j \in (0,5]$.

II. Analysis

A. *Showing Off or Laying Low?*

We now study the newcomer's incentives to signal his type. Incurring signaling costs may be a worthwhile endeavor if it transmits information to the incumbent, and this information transmission leads to outcomes which the newcomer considers beneficial. Two observations are central to understanding signaling incentives. First, under complete information, the effort $x_i^{CI}(c_{-i})$ spent by a player with marginal cost c_i against an opponent with marginal cost c_{-i} in equilibrium is maximal when she faces another player of the same talent, $\bar{x}^{CI} = x_i^{CI}(c_i)$, and decreases monotonically when the talent of the opponent increases or decreases. This is shown in Figure 2. Therefore, with potential opponent types drawn from some interval $[c_L, c_H]$, the incumbent's minimum effort under complete information will be chosen either against c_L or against c_H, and thus $\underline{x}^{CI} = \min\{x_I^{CI}(c_L), x_I^{CI}(c_H)\}$. Second, under asymmetric information, effort of a player who faces an opponent drawn from $[c_L, c_H]$ is $x^{AI} \in (\underline{x}^{CI}, \bar{x}^{CI})$ for any non-degenerate type distribution (see Lemma 4 in Appendix E). That is, the incumbent's asymmetric equilibrium effort always fulfills $x_I^{AI} \in (\underline{x}^{CI}, x_I^{CI}(c_I))$. The newcomer always benefits when the incumbent lowers her effort, and hence the best the newcomer can do is to convince the incumbent that he is the type against which she is least aggressive, i.e., the type that induces $\underline{x}^{CI}$. This is true for *all* newcomer types. Thus, our first task is to identify this type that induces $\underline{x}^{CI}$.

We can see from Figure 2 that complete information best responses are hump-shaped in the situations that we study and thus every positive complete information equilibrium effort choice—except the maximum—can be induced by two different types of opponents. Complete information effort x_i against an opponent with marginal cost c_j is $x_i(c_j) = c_j/(c_i + c_j)^2$, implying that effort is identical against

some c_j' and c_j'', $c_j' \geq c_i \geq c_j''$, if and only if $c_i = \sqrt{c_j' c_j''}$. Moreover, the closer c_j' and c_j'' to c_i, the more i spends against both. Consequently, if $c_I = \sqrt{c_L c_H} := \mu_N$, the incumbent chooses identical effort against the most and against the least talented newcomer, and thus all newcomer types are indifferent between making the incumbent believe to be either one (absent signaling costs). If the incumbent is relatively untalented, $c_I > \mu_N$, the incumbent's lowest complete information effort is against the most talented newcomer c_L, and otherwise against the least talented newcomer c_H. Thus, we have a condition that tells us—depending on just c_I, c_L, and c_H—what the newcomer would like the incumbent to believe.

Note that incentives to signal one's type are different from incentives in standard signaling games à la Spence (1973), where incentives to imitate run in one direction—worse types want to be perceived as better types and thus better types signal to prevent this. In our setting incentives for the newcomer can run in either direction, depending on the incumbent's ability. Thus, the same type distribution and preferences do not, by themselves, determine the direction of signaling incentives. The reason for this is that the fierceness of subsequent competition depends on the newcomer's as well as the incumbent's ability.

The implications of the described forces for signaling are most easily seen when there are only two types of newcomers. In this case, if one type of newcomer successfully signals his type, the other type is revealed as well. By the arguments above, if $c_I < \mu_N$ and thus the incumbent is relatively talented, both types would like to signal to have low talent. Both would like to signal to be able otherwise. Because signaling cost are correlated with talent, whenever both would like to show that they are untalented, lower signaling costs allow the more talented newcomer type to effectively mimic the less able one, and thus no separating equilibrium exists. In this situation no signaling takes place. To the contrary, when the incumbent has low talent, both types would like to signal to be talented. But now the more talented newcomer type can induce a separating equilibrium exactly because he has lower costs of signaling, and in equilibrium all types will be revealed. Hence, our first conclusion, based on the binary case, is that if the newcomer is sufficiently talented, information on the newcomer's talent is fully revealed through signaling in equilibrium.[17]

PROPOSITION 1: *Assume binary types and let $\mu_N := \sqrt{c_L c_H}$. If the incumbent is relatively weak, $c_I > \mu_N$, a unique equilibrium exists in which the newcomer's type is completely revealed. Otherwise, no information will be revealed in equilibrium.*

The proposition illustrates that it does not suffice to be just more talented than the incumbent for the talented newcomer to use signaling to discourage or "psych out" the incumbent (i.e., $c_L < c_I$); the advantage needs to be sufficiently

[17] Note that Proposition 1 and all results that follow pertaining to the binary case locally generalize to a more general class of contest success functions of the form $p_i = f(x_i)/\big(f(x_N) + f(x_I)\big)$, where $f(x_i)$ is an increasing and weakly concave function. The prerequisite for this is a critical value of c_I, which generally exists and is characterized by the equality of the incumbent's full information efforts against c_H and c_L. Details are available from the authors upon request.

large for any given c_H. This can be seen easily from the condition for signaling to happen: $c_I > \sqrt{c_L c_H} \Leftrightarrow c_L < c_I^2/c_H < c_I$. A non-exceptional newcomer prefers to send no signal at all, as revealing his talent would only sharpen competition.[18]

Note that equilibrium signaling in Proposition 1 does not depend on the particular distribution of newcomer types. This is a peculiarity of the binary case, because, in this case, when the talented newcomer signals his type, the untalented newcomer's type is also revealed. When there are at least three types of newcomers, the distribution of types matters. For example, assume there is a third newcomer type $c_M = c_I \in (c_L, c_H)$. The above reasoning remains valid: If the incumbent is strong, $c_I < \mu_N$, all types would like to signal to have low talent, c_H. Otherwise high talent, c_L, induces the lowest incumbent effort under complete information. However, unlike in the binary scenario, there is now a third type that nobody ever wants to mimic, c_M. The reason is that the incumbent's effort is *maximized* when she is certain to face an equally talented newcomer. Consequently, c_M always has an incentive to conceal his type by pretending to have either low or high talent. Because $c_M < c_H$, pretending to have low talent is always feasible, and thus neither c_M nor c_H ever separate in equilibrium. Therefore, there are only two possible constellations in a PBE: all newcomer types refrain from signaling and no information is revealed, or only the high talent newcomer's type is revealed, while c_M and c_H pool. If $c_I > \mu_N$, the incumbent chooses the lowest effort under complete information against the high ability newcomer, and thus he will separate in equilibrium. However, if $c_I \leq \mu_N$, whether or not the high ability newcomer signals and reveal his type in equilibrium depends on the details of the type distribution. For example, assume that $c_L = 1$, $c_I = 13/10$, and $c_H = 2$, implying $\mu_N = \sqrt{2} > c_I$. Thus, whether the high ability newcomer signals his type in equilibrium in this case depends on the type distribution. If the probability of a high and low talent newcomer is equal, i.e., if $q_L = (1/2) - (1/2)q_M$ for $q_M \in (0,1)$, then the high ability newcomer separates in equilibrium if the probability of a medium talent newcomer is sufficiently large, $q_M > \bar{q}_M \approx 0.5136$. To the contrary, if $q_M \leq \bar{q}_M$, then no newcomer signals in equilibrium and thus no information is revealed. Intuitively, the higher q_M, the higher is incumbent effort after observing no signal and thus the less attractive is pooling for the talented type.

With more than just two types, *partial* separating equilibria are generally possible. In such a situation, a positive signaling effort convinces the incumbent of the signaling newcomer's talent, but, unlike in the binary case, observing no signal leaves open several levels of newcomer talent. How the incumbent reacts to the newcomer not sending a signal depends now on the distribution of types, and this in turn determines signaling incentives.

Now we turn to general distributions. The ternary example we discussed above suggests that equilibrium is monotone in the newcomer's type in the sense that if a newcomer's talent is greater than some threshold value he signals, while he refrains

[18] If we assume that there are n incumbents, incentives for signaling drastically decrease, if all these incumbents are equally talented. In particular, being perceived as the most talented newcomer, c_L is beneficial if and only if $nc_I < \sqrt{c_H c_L}$. This never holds under Assumption 1 for $n > 1$ incumbents, and hence there are no incentives to signal. However, typically not all incumbents are equally talented and the relevant competition is between a smaller peer group of similarly talented competitors only.

from signaling otherwise. We now formally define such threshold equilibria and show that equilibrium signaling is indeed monotone in c_N.

DEFINITION 1 (Threshold Equilibrium): *A threshold equilibrium is a Perfect Bayesian Equilibrium in which a threshold c_Z exists such that all newcomers with $c_N < c_Z$ (if any) optimally separate through costly signaling $s_N(c_N) > 0$, $s'_N(c_N) < 0$, while all others refrain from signaling and thus pool. The incumbent observes the newcomer's signaling choice and updates her beliefs accordingly. Both players choose efforts in the contest given these beliefs.*

Then the following is true.

PROPOSITION 2: *Any PBE exhibits signaling choices weakly monotone in the newcomer's type, implying any PBE is a threshold equilibrium. Any PBE consistent with D1 is additionally continuous in the newcomer's type.*

D1 is not necessary to generate monotone—or threshold—equilibria. However, without D1 there is potentially a continuum of such monotone PBE. D1 selects the equilibrium in which the greatest possible fraction of newcomer types reveal their talent. Thus, we restrict attention to PBE that satisfy D1 in the following.

We now state the main result from this section.

THEOREM 1: *Suppose that newcomer types are continuously distributed on $\mathcal{S}_N$ according to $F(c_N)$. There exists a unique equilibrium characterized by an equilibrium threshold incumbent ability $\tilde{c}_I > c_L$ such that*

- *if $c_I > \tilde{c}_I$, a unique threshold newcomer type $\tilde{c}_N(c_I) \in \left(c_I^2/c_H, c_I\right)$ exists. Newcomer types c_N reveal their type through costly signaling if and only if $c_N < \tilde{c}_N$. The threshold $\tilde{c}_N(c_I)$ strictly increases in c_I for $c_I \in (\tilde{c}_I, c_H]$.*
- *if $c_I \leq \tilde{c}_I$ all newcomers pool in equilibrium.*

$\tilde{c}_I$ is implicitly defined by $\tilde{c}_N(c_I) = c_L$.

Theorem 1 shows that the impact of subsequent competition on signaling incentives holds in general. Especially noteworthy is that these incentives crucially depend on the incumbent's ability—a sufficiently talented incumbent, $c_I \leq \tilde{c}_I$, destroys incentives for talented newcomers to signal, because instead of discouraging the incumbent it provokes more intense competition.

Figure 3 presents a graphical illustration of Theorem 1 when the newcomer's ability is uniformly distributed on $[1,2]$. The solid line represents the equilibrium newcomer type threshold $\tilde{c}_N$ (N-threshold in the following), which increases in c_I. Newcomer types lying below this threshold reveal their types through the costly signal, those lying above do not. When the incumbent is talented, that is when c_I lies to the left of the dotted vertical line ($\tilde{c}_I$), all newcomer types lie above the threshold and no one reveals their type. As c_I increases, so too does the amount of types that reveal their type. Indeed, an incumbent corresponding to the worst

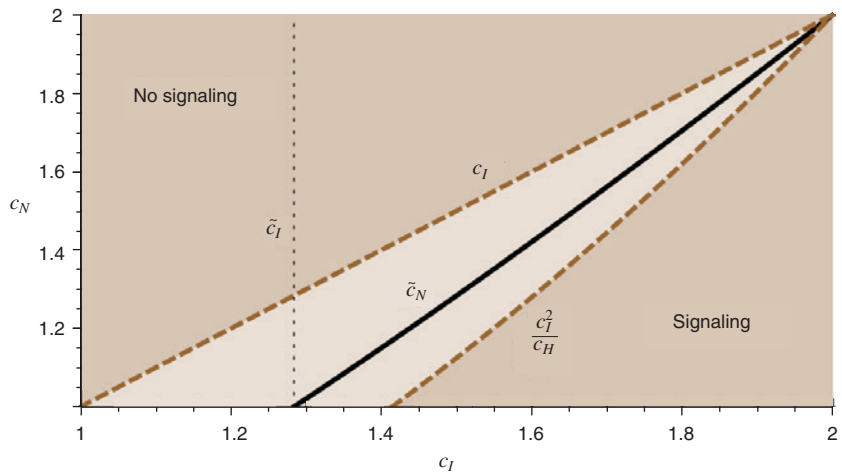

FIGURE 3

Notes: The graphic show the N-threshold (solid line) when $c_N \sim U[1,2]$. The dashed lines represent the bounds of the N-threshold. All newcomer types below the lower bound always separate in equilibrium independent of the type distribution's details. Similarly, all newcomer types above the upper bound always pool. The dotted vertical line represents $\tilde{c}_I$.

possible newcomer type leads to a complete separating equilibrium. Figure 3 also displays the upper and lower bounds of the N-threshold, represented by the dashed lines, which are not distribution specific. Thus, regardless of the type distribution, newcomer types lying below the lower bound always reveal their type in equilibrium.

One might wonder whether there is a systematic relation between the distribution of newcomer types and the incentives to signal. No easy general relationship exists, as is most easily seen in the binary case. There the geometric mean of newcomer types determines the signaling threshold, i.e., signaling occurs if and only if the incumbent's type lies above the threshold. The key thing is to notice that this threshold is independent of the probabilities of the newcomer's type. As a consequence, a change in the distribution may increase or decrease the threshold, depending on how the geometric mean is affected. More generally, only the probability distribution of newcomer types in the pooling segment and the range of the complete probability distribution impact the threshold.

At this point it is useful to link our findings to the ones in Katsenos (2009). He studies a two-sided asymmetric information framework where ability distributions of both competitors are the same, i.e., either competitor is of type c_L with probability q and type c_H with probability $1 - q$. He finds that a separating equilibrium only exists if a player's opponent is sufficiently weak in expectation, i.e., when q is sufficiently small. Thus, also with two-sided asymmetric information a variant of our result applies: Only a sufficiently talented competitor reveals his type in equilibrium.

B. *The Incumbent's Perspective*

One might be tempted to think that type revelation by the newcomer must come at a cost to the incumbent. The underlying intuition suggests that signaling discourages the incumbent, thus reducing her chance of winning the contest, and hence making her worse off. The flaw in this intuition is that it ignores how the newcomer responds to the incumbent's change in effort.

As in the previous section, we start with the binary case to develop intuition. Absent signaling, the incumbent remains uncertain as to the newcomer's ability and hence chooses effort somewhere between that which she would choose knowing the newcomer is a low type and that when knowing he is a high type, i.e., $x_I^{AI} \in \left[\min\left\{x_I^{CI}(c_L), x_I^{CI}(c_H)\right\}, \max\left\{x_I^{CI}(c_L), x_I^{CI}(c_H)\right\}\right]$. As we next show, whether signaling harms the incumbent depends on the relationship of this intermediate effort to what an incumbent with commitment power would choose under full information.

To understand this, consider the following hypothetical problem: What effort level would the incumbent choose, if she could commit to *any* effort level she wishes to? Such a situation is alike a Stackelberg game with the incumbent as the leader and the newcomer as the follower as well as full information regarding the newcomer's talent. Under full information, it is well-known that a Stackelberg leader incumbent would *overinvest* in effort against an untalented newcomer and *underinvest* against a talented one (see, for example, Dixit 1987). The reason is that weaker opponents are deterred by greater effort levels and stronger opponents are appeased by lower ones. This follows from the hump-shaped best response functions that we have seen in Figure 2. Furthermore, in this setting the incumbent's utility is single-peaked in her own action and is maximized when the appropriate Stackelberg effort is chosen (see Yildirim 2005). Whether the incumbent's efforts under type revelation or under ignorance lie closer to the Stackelberg efforts depends crucially on the incumbent's relative strength compared to the newcomer.

The upper graph of Figure 4 depicts a situation where the incumbent is relatively weak, i.e., $c_I > \mu_N$. Notice that, in this circumstance, she exerts less effort against a talented newcomer and more against an untalented one. With commitment, her ideal effort is even lower versus a talented newcomer and higher in the opposite case, as shown by points $x_I^{SB}(c_L)$ and $x_I^{SB}(c_H)$ in the figure. Absent signaling, the incumbent optimally chooses a point between the full information efforts $x_I^{CI}(c_L)$ and $x_I^{CI}(c_H)$ in the figure. Since this places her further from the optimal commitment strategy regardless of the newcomer's type, the incumbent benefits when the newcomer signals.

The situation where the incumbent is relatively strong, $c_I < \mu_N$, is more complicated. The ordering of the complete information efforts reverses and hence the Stackelberg actions may lie outside or inside their range. The lower graph of Figure 4 depicts the former. Clearly the incumbent's effort under ignorance is closer to her optimal commitment strategy regardless of the newcomer's type. In the latter case the ranking is not as clearly visible and ignorance may only be optimal in expectation. Fortunately, the condition for ignorance to be bliss is quite simple.

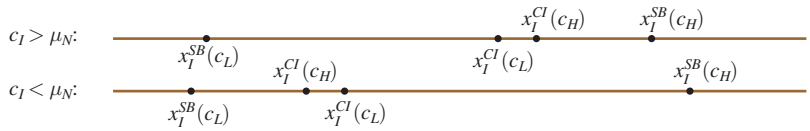

FIGURE 4

Notes: Efforts rankings for the incumbent in the complete information Nash equilibrium and when she is a Stackelberg leader against the different newcomer types. In the upper graph we have $c_I > \mu_N$, implying $x_I^{CI}(c_L) < x_I^{CI}(c_H)$, and thus remaining ignorant moves the incumbent farther away from the Stackelberg points vis a vis both newcomer types. Thus, the incumbent benefits from signaling. In the lower graph we have $c_I < \mu_N$, implying $x_I^{CI}(c_L) > x_I^{CI}(c_H)$, and thus remaining ignorant moves the incumbent closer to the Stackelberg points versus both newcomer types. Hence, ignorance is bliss for the incumbent.

PROPOSITION 3: *Consider binary types. If the incumbent is relatively talented compared to the newcomer, $c_I < \mu_N$, she strictly prefers to stay ignorant about the newcomer's type. If the incumbent is relatively untalented, i.e., when $c_I > \mu_N$, she prefers to obtain information about the newcomer. When $c_I = \mu_N$, the incumbent is indifferent between obtaining information and staying ignorant.*

Depending on the talent of the incumbent, she may both prefer the newcomer to signal and thus to reveal his type or to remain opaque by choosing not to signal. Note that the condition delineating her preferences between these two informational scenarios is equivalent to the one in Proposition 1 governing signaling: If the newcomer signals, the incumbent wants the newcomer to signal; if the newcomer does not reveal his talent, the incumbent does not want to learn the newcomer's talent either. Their interests regarding information revelation are perfectly aligned.

Now consider the general set-up. With at least three types, as we showed earlier, we may have partial separation, and thus some newcomer types separate through signaling while others do not. We now study how the incumbent likes the possibility of signaling *in expectation*. In particular, we consider a thought experiment. Suppose that the incumbent could pick any threshold newcomer type $c_Z \in [c_L, c_H]$. All types with lower costs reveal their types, and all types with higher costs refrain from doing so. Which threshold would she choose? Proposition 4 offers an answer.

PROPOSITION 4: *Among all possible thresholds, the N-threshold $\tilde{c}_N$ maximizes the incumbent's expected utility.*

Proposition 4 shows that, surprisingly, the threshold most preferred by the incumbent perfectly coincides with that "chosen" by the newcomer. Thus, despite their conflict over receiving the reward, there is no conflict as regards the tactics of signaling.

We can find back the intuition for the binary case also in this more general case, focusing on changes locally around the equilibrium threshold $\tilde{c}_N$. In Lemma 5 in the Appendix we show that the efforts the incumbent chooses under complete information against the newcomer at the N-threshold $\tilde{c}_N$ and against the pooling newcomers are identical. Furthermore, when the signaling threshold increases above $\tilde{c}_N$, the incumbent chooses greater effort against the threshold type than against the pooling

newcomers, while the effort ranking reverses when the threshold decreases below the N-threshold.

When a threshold marginally lower than $\tilde{c}_N$ is chosen, some newcomers with type $c_N < \tilde{c}_N$ that used to reveal their types now refrain from doing so. Against these newcomers the incumbent now chooses a level of effort that is greater than it used to be when they were still signaling. However, our discussion leading to 3 revealed that against these newcomers the incumbent would optimally *lower* efforts to appease them, because they have greater talent than the incumbent. At the same time, the effort chosen against the pooling newcomers remains roughly constant, and thus has only second order utility consequences. Hence, a lower threshold is not in the interest of the incumbent.

The same logic remains true when a threshold marginally higher than $\tilde{c}_N$ is chosen. In this case some newcomers with types $c_N > \tilde{c}_N$ that used to pool with the low ability newcomers now reveal their types. Against these newcomers the incumbent now also chooses a level of effort that is greater than before. Also against these newcomers the incumbent would optimally choose *lower* efforts to appease them. Meanwhile, the effort chosen against the pooling newcomers changes only marginally, which has negligible implications for the incumbent's expected utility. Hence, a marginally lower threshold is not in the interest of the incumbent, either.

III. Cutting Down the Tall Poppies?

Our previous analysis reveals that both parties agree about the talent threshold at which signaling—or showing off—improves matters. Yet societal norms are not so careful in accepting such behavior—punishment is not conditional on talent. In Australia and New Zealand the distaste for showing off is described as "cutting down the tall poppies," while in Scandinavia the "Law of Jante" represents the cultural proscription against showing off. In Japan the social obligation to conform and the consequences for those who violate this norm are described in a common proverb: "The nail that sticks out shall be hammered down."[19] On the other hand, American cultural norms, particularly in business contexts, have few such proscriptions.[20] What constitutes a social norm generally depends on local cultures and thus might differ significantly from one society to the next (Ostrom 2000). Behavior that is acceptable in one society might be considered deviant in another. One way to explain such diversity in norms is the existence of a "horses for courses" effect: while under some circumstances society as a whole benefits from a given norm—for example, from a norm discouraging showing off—under other circumstances the same norm has detrimental effects. In this section we endeavor to provide a ratio-

[19] Okada, Rokuo. 1955. *Japanese Proverbs and Proverbial Phrases*.Tokyo: Japan Travel Bureau, page 28.

[20] For example, Fineman (2015) emphasizes the importance of showing off:

> The other side of the coin is this: bragging is integral to career now. You can denounce it, or you can learn how to use it to your advantage. At least, that's what I've done. Or rather, what I teach people to do. How to leverage voice and know-how and learn how to break through the noise. Or in more simple terms, redefine self-promotion to get ahead in the workplace. It's the truth of 2015, and it matters at every phase of your career.

nale for the diversity in norms against showing off. To this end we need to adapt our model to account for such proscriptions. For social norms to exist it is important that deviant, norm violating behavior will be sanctioned (Akerlof 1980, Ostrom 2000). Sanctioning makes it costly to violate a norm, and the stricter a norm is, the costlier is norm violation.

In the following we thus assume that due to a norm, exerting positive signaling effort implies the newcomer incurs an additional fixed cost $\xi \geq 0$. Formally, we assume that signaling costs now become $c_N s_N + \xi$ for any $s_N > 0$.

First consider a setting where the efforts expended in the contest are socially desirable as, for example, in a school setting, where the contest incentivizes human capital formation, or a workplace setting, where the contest incentivizes managers' efforts. If one considers expected aggregate effort as reflecting the benefits to society our model offers a crisp answer to the question as to whether norms that discourage signaling are helpful or harmful.

THEOREM 2 (Norms and Aggregate Effort): *A higher fixed cost ξ of signaling monotonically decreases the equilibrium threshold and thus the fraction of newcomers revealing their type. Furthermore, expected aggregate effort increases as the threshold for signaling decreases relative to the N-threshold $\tilde{c}_N$, i.e., when there is less signaling.*

While Proposition 4 showed that there was complete alignment between competitors as to the optimal information disclosure regime, their interests do not align as neatly with those of a society interested in aggregate effort. In fact, the degree of showing off that occurs in equilibrium represents society's *worst* information disclosure regime. Any norm that discourages some newcomer types from signaling increases expected aggregate effort, but if possible, society would gladly suppress signaling behavior entirely. Of course, no model can fully reflect the rich social milieu the players inhabit. Nonetheless, Theorem 2 offers useful intuitions and insights. It shows that the global norm of punishing show-offs may not be dysfunctional cultural baggage from the past in need of discarding. Instead, it could represent a sensible societal choice.[21]

But does it also suggest that US firms that embrace a "culture of showing off" have somehow gotten it wrong, that they are pursuing a policy that makes them worse off? In rigid promotion systems where seniority mainly determines managerial responsibility, and where contest structures are used predominantly to determine bonus payments, as often the case in Japanese firms, aggregate effort seems the most important consideration.[22] Likewise, in schools, where educating everyone, rather than sifting out talent, is the main goal, effort once again seems like the appropriate welfare metric.

[21] Theorem 2 also applies to settings where effort is socially wasteful such as the competition for political rents. In these settings, consequently, welfare decreases in ξ.

[22] For example, Chlosta et al. (2014) report that, unlike in the United States, contest structures in Japan are indeed predominantly used to determine bonus payments, suggesting that the main goal of the contest is to incentivize effort.

But in other business contexts, this narrow view of welfare often seems more problematic. Businesses thrive by identifying rare talent and exploiting that for competitive advantage. This suggests that selection ought to have considerable weight in assessing the success of a business culture. Likewise, when success depends primarily on innovation, as with many Silicon Valley firms, effort cannot be the sole consideration. If selection efficiency is measured by the expected talent c_i of the winner of the contest, a sharp result is available in the polar case, i.e., where only efficient selection matters.

THEOREM 3 (Norms and Selection): *A higher fixed cost ξ of signaling monotonically decreases the fraction of newcomers revealing their type and thus the equilibrium threshold. Furthermore, the expected talent of the winner decreases as the threshold for signaling decreases relative to the N-threshold $\tilde{c}_N$, i.e., when there is less signaling.*

Norms that sanction signaling are conducive to increasing aggregate effort. But as Theorem 3 shows, this comes at a price: it deteriorates selection efficiency. The reason is that signaling, when it happens, reveals the newcomer's superior talent, which discourages the incumbent and thus lowers her effort. However, by accentuating the competitors' differences, it also leads to greater differences in their behavior, which benefits selection efficiency. Thus, it may be that US firms are not erring by embracing such a laissez-faire culture. Rather, differences in business culture might simply reflect differing tradeoffs between effort and selection.

At this point it is worthwhile to briefly discuss the importance of the D1 refinement for the validity of Theorems 2 and 3. As highlighted in Proposition 2, all PBE exhibit signaling behavior weakly monotone in the newcomer's type, and an implication of the proposition is that there is no PBE with more newcomer types signaling than in the D1 equilibrium. This implies that Theorems 2 and 3 remain valid in any PBE that does not satisfy D1.

In many settings, as, for example, in the workplace, newcomers themselves may be biased in their perception of their own talent (see e.g., Malmendier and Tate 2005, 2015) for evidence of overconfidence in the corporate world and Ortoleva and Snowberg (2015) for evidence of overconfidence in the political arena). A prevalence of overconfident newcomers might dictate against a norm against signaling. The reason is that when the newcomer is overconfident, signaling provides an additional benefit—it gives feedback to the newcomer as to his true ability. Remember that only newcomers who believe they are exceptional signal. An overconfident newcomer, who believes he is exceptional but is, in fact, quite ordinary, will be disabused of this notion when his attempts at signaling his great talent fall flat. This realization causes him to step up effort, as talent alone will not suffice. Also the incumbent increases her effort in response. Of course, we know from the previous analysis that this feedback comes at a cost—when the newcomer is, in fact, exceptional, signaling reveals this and leads to effort-reducing discouragement and complacency effects as described above. Thus, there is a tradeoff: when the newcomer is likely to be well calibrated as to his ability, a norm condemning signaling

is optimal; whereas, when overconfidence is rife, a culture of signaling is optimal. We show this formally for the binary case.

PROPOSITION 5: *With binary types, if the probability of an overconfident newcomer is large, expected aggregate effort decreases when signaling is successfully discouraged.*

Appendix J adds overconfidence to the model with binary types and formalizes the intuition above about the feedback benefit of signaling.

IV. Productive Signals

In our preceding analysis, we upheld the assumption that signaling is not *directly* productive but only facilitates information transmission from the newcomer to the incumbent. Yet, in some situations, signaling might have a direct influence on the contest outcome. In the context of our model, this implies that signaling, or some fraction κ of the signaling effort, translates into productive effort in the contest.[23] How does this influence our results?

Recall that the main motive for (unproductive) signaling is influencing the incumbent's beliefs in such a way that she reacts with low effort in the contest. When signaling is productive, adding to productive effort at rate κ, then two new motives appear. First, since the newcomer now has access to two "channels" for exerting effort, he will need to choose these optimally given his type. Second, even if signaling is not an especially efficient effort producing technology, the newcomer may be motivated to employ it so as to gain a commitment advantage. If the newcomer is talented, he is stronger than the incumbent, and that gives him an incentive to commit to a higher effort relative to the Nash equilibrium to soften competition; if he is untalented, he would like to commit to lower effort with the same intention (see Dixit 1987). While commitment to a higher effort level is generally possible, commitment to a lower effort level is not, because the newcomer has a second chance to choose effort and he would always increase his effort to match the complete information Nash equilibrium effort (see Lemma 1 of Yildirim 2005). Thus, when the signal is productive, a strong newcomer now has two ways to affect competition. He can signal his type through signaling and thereby alter effort responses in the second stage game. He can also use the effort produced by signaling as a commitment device, transforming the situation to one resembling Stackelberg competition. Both channels can be used to discourage the incumbent. When κ is small, the commitment channel is too expensive and our original analysis still holds. When κ is large enough, commitment motives dominate.

[23] Essentially, this implies that the newcomer can invest early and late in the contest, with a potentially differing effectiveness or effective costs of effort. There are various reasons to assume the productivity or cost of effort differs across time, including learning by doing (Clark and Nilssen 2013), recency bias (Fudenberg and Levine 2014, 2016), or sabotage of early stage efforts (Chowdhury and Gürtler 2015).

PROPOSITION 6: *When types are binary and signaling produces productive effort at rate κ per unit, then Theorems 1 to 3 hold so long as $\kappa < \bar{\kappa} = 2/3$. In particular, a sufficiently high fixed cost of signaling yields higher aggregate effort, while selection is inhibited.*

The threshold where productive effort starts affecting our results highlights the relative importance of the two new factors introduced by this extension. Notice that, if the allocation of productive effort across channels were the main driver, then the relevant threshold would be $\kappa = 1$. When $\kappa > 1$, signaling is the more efficient effort producing technology, whereas when $\kappa < 1$, it is inefficient. Instead, the critical value of κ represents a threshold for commitment forces to come into play. In a sense, κ characterizes the expense of a commitment strategy—commitment is cheaper for higher values of κ. Since no such commitment was present in the baseline model, it then follows that, so long as commitment via signaling remains unattractive, the strategic considerations of the two parties are unchanged. By contrast, when commitment via signaling is sufficiently cost effective, the (talented) newcomer avails herself of this strategy and the game becomes, in effect, sequential with the newcomer having the first move in generating effort via signaling.[24]

V. Conclusion

The Cambridge dictionary defines "showing off" as "to behave in a way that is intended to attract attention or admiration, and that other people often find annoying."[25] This definition, especially the latter part, is in line with social norms around the globe, for example the Tall Poppy Syndrome known in large parts of the English speaking world or the Law of Jante known in Nordic countries. At the same time, in many business environments, as for example in Silicon Valley, no such proscriptions exist, and showing off one's talent is often even well regarded. In this paper we study the economics of showing off and laying low and its consequences. We augment a standard contest model by allowing a new competitor (newcomer) to transmit information through showing off (costly signaling) to an established competitor (incumbent). Our analysis reveals that the desirability of showing off is indeed a question of horses for courses.

Costly signaling allows a contestant to influence the beliefs of his contender. We found that, unlike in standard signaling games à la Spence (1973), incentives to signal one's type to a competitor before a contest can run in either direction: a weak contender might want to mimic a strong one, and a strong contender might want to mimic a weak contender, in both cases with the aim to decrease the fierceness of ensuing competition. But while meager talent is easily mimicked, it is hard to mimic someone

[24] Note that giving the incumbent a chance to "show off" as well, i.e., to expend productive effort at rate κ ahead of the contest does not change our findings qualitatively when κ is sufficiently low. The reason is that the incumbent benefits from the newcomer's signaling and prefers to use the information the newcomer provides him with instead of tying her hands by pre-committing to a high effort level.

[25] https://dictionary.cambridge.org/dictionary/english/show-off. Interestingly the American definition merely states: "to do something to attract attention to yourself" and thus does not come with a negative connotation.

exceptional, and hence only the exceptional will successfully engage in showing off and reveal their type in equilibrium, enabling them to psych out their competition.

While the newcomer decides on whether to send a costly signal, the incumbent may not have a choice in this matter. Surprisingly, though, the incumbent is well-served by the signaling choices of the newcomer. While showing off aims at psyching out the opponent, who reacts by reducing effort, it also decreases the general fierceness of competition. When showing off happens, it leads to an overall reduction in effort. In this sense, pre-conflict showing off behavior can be seen as a form of collusion, benefitting both competitors. Our analysis thus offers a rationale for the emergence of norms against showing off in circumstances where societies' concerns lie with aggregate effort.

On the other hand, in settings where selection of the most talented is paramount, a norm against showing off is not advisable. Showing off tends to accentuate competitors' differences and thus helps selecting the best. Because innovative firms, like startups, rely exactly on that, our results shed some light on why a norm against showing off seems absent from Silicon Valley business etiquette.

Mathematical Appendix

Appendix A. Preliminaries: Equilibrium under Full and Asymmetric Information

In this section we derive equilibrium efforts as well as utility of the subgame where all types of newcomers from the interval $[c_L, c_Z]$ reveal their types, while newcomers in $[c_Z, c_H]$ do not reveal their types. In both cases, the incumbent updates her beliefs accordingly. Note that this analysis nests the full information case when $c_Z = c_H$ (i.e., all newcomers reveal their types) as well as the no information case when $c_Z = c_L$ (i.e., no newcomer reveals his type). Also note that other authors have derived these latter efforts before, see Nti (1999) for the complete information case ($c_Z = c_H$) and Hurley and Shogren (1998) for the asymmetric information case ($c_Z = c_L$). The latter express efforts as a function of players' "relative resolve" and "willingness to waste," which is not ideal for our purpose. Hence, we derive efforts directly as a function of the newcomer's type distribution in the following.

LEMMA 1: *Let c_N be distributed according to some distribution $F(c_N)$ over the interval $[c_L, c_H]$ with strictly positive density $f(c_N)$. Moreover, let $c_H \leq 4c_L$. Assume all newcomer types $c_N < c_Z$ signal their type, while all others abstain and pool. The subgame in which the incumbent does not receive a signal has a unique equilibrium, which is in pure strategies:*

$$x_I^{AI} = \frac{\left(\int_{c_Z}^{c_H} \sqrt{c_N}\, dF(c_N)\right)^2}{\left(c_I\left(1 - F(c_Z)\right) + \int_{c_Z}^{c_H} c_N\, dF(c_N)\right)^2} > 0$$

and

$$x_N^{AI} = \sqrt{\frac{x_I^{AI}}{c_N}} - x_I^{AI} > 0.$$

Ex post, the corresponding equilibrium utility of the incumbent is

$$\pi_I^{AI} = \frac{x_I^{AI}}{x_I^{AI} + x_N^{AI}} - c_I x_I^{AI} = \sqrt{c_N x_I^{AI}} - c_I x_I^{AI}.$$

In case the incumbent learns the newcomer's type, the subgame equilibrium follows from the equilibrium characterization under complete information:

$$x_i^{CI} = \frac{c_j}{(c_I + c_N)^2}, \quad X^{CI} \equiv x_I^{CI} + x_N^{CI} = \frac{1}{c_I + c_N}, \quad \pi_i^{CI} = \frac{c_j^2}{(c_I + c_N)^2}.$$

It is easily verified that the incumbent will invest more against a low-cost opponent c_L than against a high-cost one c_H if and only if $c_I < \sqrt{c_H c_L}$.

Ex ante the utility of the incumbent when all newcomer types in the interval $[c_L, c_Z]$ signal and thus reveal their types equals

$$E\pi_I = \int_{c_L}^{c_Z} \frac{c_N^2}{(c_N + c_I)^2} dF(c_N) + \frac{\left(\int_{c_Z}^{c_H} \sqrt{c_N}\, dF(c_N)\right)^2 \int_{c_Z}^{c_H} c_N dF(c_N)}{\left(c_I\left(1 - F(c_Z)\right) + \int_{c_Z}^{c_H} c_N dF(c_N)\right)^2}.$$

Similarly, expected aggregate effort is equal to

$$EX = \int_{c_L}^{c_Z} \frac{1}{c_N + c_I} dF(c_N) + \frac{\int_{c_Z}^{c_H} \sqrt{c_N}\, dF(c_N) \int_{c_Z}^{c_H} \frac{1}{\sqrt{c_N}} dF(c_N)}{\left(c_I\left(1 - F(c_Z)\right) + \int_{c_Z}^{c_H} c_N dF(c_N)\right)}.$$

PROOF:

The incomplete information (sub-)game can be solved easiest by taking the FOC of the incumbent

$$\frac{1}{1 - F(c_Z)} \int_{c_Z}^{c_H} \frac{x_N(c_N)}{\left(x_I + x_N(c_N)\right)^2} dF(c_N) = c_I$$

and then inserting the best-response of all newcomer types, $BR_N(x_I; c_N) = \max\left\{0, \sqrt{x_I / c_N} - x_I\right\} = \sqrt{x_I / c_N} - x_I$ when $4c_L \geq c_H$, which is guaranteed by Assumption 1. This yields

$$\frac{1}{1 - F(c_Z)} \int_{c_Z}^{c_H} \frac{\sqrt{\frac{x_I}{c_N}} - x_I}{\left(x_I + \sqrt{\frac{x_I}{c_N}} - x_I\right)^2} dF(c_N) = c_I$$

$$\Leftrightarrow \int_{c_Z}^{c_H} c_N \left(\frac{1}{\sqrt{x_I c_N}} - 1\right) dF(c_N) = c_I\left(1 - F(c_Z)\right)$$

$$\Leftrightarrow \frac{\int_{c_Z}^{c_H} \sqrt{c_N}\, dF(c_N)}{\sqrt{x_I}} = c_I\left(1 - F(c_Z)\right) + \int_{c_Z}^{c_H} c_N dF(c_N)$$

$$\Leftrightarrow x_I^{AI}(c_Z) = \frac{\left(\int_{c_Z}^{c_H} \sqrt{c_N}\, dF(c_N)\right)^2}{\left(c_I\left(1 - F(c_Z)\right) + \int_{c_Z}^{c_H} c_N dF(c_N)\right)^2}.$$

The corresponding SOC is fulfilled, as

$$-\frac{1}{1-F(c_Z)}\int_{c_Z}^{c_H}\frac{x_N(c_N)}{\left(x_I+x_N(c_N)\right)^3}dF(c_N) \;<\; 0,$$

and thus the FOC characterize a global utility maximum. Similar steps establish a global maximum for the newcomer. Expected utility and expected aggregate effort follow directly from the equilibrium effort levels. ∎

Appendix B. Preliminaries: Separating and Pooling Equilibria

In this section we collect a series of results that will help to establish the proofs of Proposition 1, Proposition 2, and Theorem 1.

Separating Equilibria.—A newcomer's maximized utility, when his opponent believes him to be of type c' after observing a signal $s(c')$, while in fact he is of type c_N, is

$$\text{(B1)}\qquad \pi_N^S(c';c_N) \;=\; \frac{BR_N\left(x_I(c');c_N\right)}{BR_N\left(x_I(c');c_N\right)+x_I(c')} - c_N BR_N\left(x_I(c');c_N\right) - c_N s(c')$$

$$= \frac{\left(c_I-\sqrt{c_N c'}+c'\right)^2}{\left(c_I+c'\right)^2} - c_N s(c'),$$

which follows from Lemma 1.

Next we prove that for any two c_N' and c_N'' such that $c_I \le \sqrt{c_N' c_N''}$, it is not possible to find a pair of signals that reveal the newcomers' types.

LEMMA 2: *Let there be a range of values $\mathcal{S}' = \left[\underline{c}_N, \overline{c}_N\right]$ such that $\underline{c}_N < \overline{c}_N$ and $c_I \le \sqrt{\underline{c}_N \overline{c}_N}$. Then, a separating equilibrium among $c_N \in \mathcal{S}'$ does not exist.*

PROOF:

Take two types from $\mathcal{S}'$, c_N' and c_N'', $c_N' < c_N''$. Using (B1), we can write the incentive constraints as follows:

$$c_N'\colon\quad \pi_N^S(c_N';c_N') \;\ge\; \pi_N^S(c_N'';c_N'),$$

$$c_N''\colon\quad \pi_N^S(c_N'';c_N'') \;\ge\; \pi_N^S(c_N';c_N''),$$

or

$$\frac{\dfrac{c_I^2}{\left(c_I+c_N'\right)^2} - \dfrac{\left(c_I-\sqrt{c_N''c_N'}+c_N''\right)^2}{\left(c_I+c_N''\right)^2}}{c_N'} \;\ge\; s(c_N') - s(c_N'')$$

$$\ge\; -\frac{\dfrac{c_I^2}{\left(c_I+c_N''\right)^2} - \dfrac{\left(c_I-\sqrt{c_N''c_N'}+c_N'\right)^2}{\left(c_I+c_N'\right)^2}}{c_N''}.$$

For separation to be possible it is necessary that

$$\frac{\dfrac{c_I^2}{(c_I+c_N')^2}-\dfrac{\left(c_I-\sqrt{c_N''c_N'}+c_N''\right)^2}{(c_I+c_N'')^2}}{c_N'} > -\frac{\dfrac{c_I^2}{(c_I+c_N'')^2}-\dfrac{\left(c_I-\sqrt{c_N''c_N'}+c_N'\right)^2}{(c_I+c_N')^2}}{c_N''},$$

which is the case if and only if $c_I > \sqrt{c_N'c_N''}$. Thus, for $c_I \leq \sqrt{\underline{c}_N, \bar{c}_N}$ there always exist some newcomer type that has an incentive and the willingness to imitate another newcomer type $c_N \in \mathcal{S}'$. In particular, all newcomer types want to and are able to imitate $\bar{c}_N$. Consequently, separation is not possible on the interval $\mathcal{S}'$. ∎

In the binary case this immediately implies when $c_I \leq \mu_N = \sqrt{c_L c_H}$ no separation is possible and newcomers need to pool. To the contrary, when $c_I > \sqrt{c_L c_H}$, separation is always possible. For the general case this implies, that there cannot be separation on the interval $\left[c_I^2/c_H, c_H\right]$, where $c_I^2/c_H < c_I$, and thus no newcomers that are less talented than the incumbent will be able to separate.

Pooling Equilibria.—The following lemma will be useful in narrowing down the set of pooling equilibria using the D1 refinement.

LEMMA 3: *Assume a candidate pooling equilibrium interval* $\left[\underline{c}_N, \bar{c}_N\right]$ *with pooling effort* $s^P \geq 0$ *and respective incumbent contest effort* x_I^P. *Consider a deviation by newcomer with talent* c_N *to some signaling effort* $s^D \geq 0$, *which causes the incumbent response* x_I^D. *Then the set of incumbent responses such that the newcomer with talent* c_N *is at least as well off after the deviation as in the pooling equilibrium is largest for type* $\underline{c}_N$ *when* $s^D > s^P$ *and for* $\bar{c}_N$ *when* $s^D < s^P$.

PROOF:

Define the incentive to deviate from any pooling equilibrium with signal $s^P \geq 0$ with accompanying incumbent effort of x_I^P as follows,

$$\text{(B2)} \quad \Delta^{Dev}(c_N) \equiv \left(1-\sqrt{c_N x_I^D}\right)^2 - c_N s^D - \left[\left(1-\sqrt{c_N x_I^P}\right)^2 - c_N s^P\right],$$

where x_I^D is the effort chosen by the incumbent after she sees a deviation to $s = s^D$. Here, $\Delta^{Dev}(c_N) = 0$ implicitly defines the upper limit $\bar{x}_I^D(c_N, s^P)$ of the range of incumbent responses such that the newcomer with talent c_N is at least as well off after the deviation as in the pooling equilibrium. First note that

$$\frac{\partial \bar{x}_I^D}{\partial\left(s^D - s^P\right)} = -\frac{\sqrt{c_N x_I^D}}{1-\sqrt{c_N x_I^D}} < 0.$$

Intuitively, if a deviation requires an increase (a decrease) in the signaling effort, the range of incumbent responses that make the deviation profitable goes down (up).

Now note that $\partial^2 \bar{x}_I^D / \partial\left(s^D - s^P\right)\partial c_N < 0$, because $x_I^D < 1/c_N$ in an interior equilibrium, as otherwise the newcomer's best response $BR_N(x_I) = \max\left\{0, \sqrt{x_I/c_N} - x_I\right\}$ is to choose zero effort. Therefore, the untalented newcomer's incentives to deviate when the extra signaling effort $s^D - s^P$ needed for the deviation increases, decrease more steeply than the able newcomer's incentives. Now let $s^P = s^D$. Then, clearly, $\bar{x}_I\left(c_N, s^D\right) = x_I^P$, which is independent of c_N. Hence, when $s^P = s^D$, the incentives to "deviate" to $s = s^D$ are the same for both, but when $s^D > s^P$, the most talented newcomer has the greatest incentive to deviate, while when $s^P > s^D$, the least talented newcomer has the greatest incentive to deviate. ∎

Appendix C. Proof of Proposition 1: The Binary Case

Assume $c_I \leq \mu_N$. By Lemma 2 we know that no separating equilibria are possible in this case. We now show that the only pooling equilibrium consistent with D1 is the efficient pooling equilibrium in which $s^P = 0$. Assume $s^P > 0$. By Lemma 3 we know that the incentives to deviate to lower signaling effort are greatest for the lowest ability type. Thus, upon observing such a deviation to a lower signal, the incumbent needs to infer that the newcomer has talent c_H and react appropriately by choosing low effort (recall that $c_I \leq \mu_N$). But then both newcomers have strict incentives to actually deviate to $s = 0$, and thus no pooling equilibrium with $s^P > 0$ is possible.

When $c_I > \mu_N$ and thus all newcomer types would like to show to have high talent, we know by Lemma 2 that there exist $\Delta_s \equiv s_L - s_H > 0$ such that both incentive constraints hold. Note that independent of the beliefs the incumbent might hold when she observes no signaling at all, the untalented newcomer has an incentive to deviate from any $s_H > 0$ by stopping signaling altogether, and thus we need $s_H = 0$ in any PBE. Further, upon observing any s' that is equilibrium dominated for H but not for L, L clearly has greater incentives to choose this level of signaling than H, and thus D1 demands that the incumbent assumes the newcomer to have high talent c_L after observing s'. For any s such that

$$s \geq \bar{s} \equiv \frac{\left(c_L + c_I - \sqrt{c_L c_H}\right)^2}{c_H\left(c_L + c_I\right)^2} - \frac{c_I^2}{c_H\left(c_H + c_I\right)^2} > 0$$

this is the case. Hence, the unique separating PBE consistent with D1 has $s_L = \bar{s}$ as well as $s_H = 0$.

Could there be a pooling equilibrium when $c_I > \mu_N$? By Lemma 3 we know that the incentives to deviate to higher signaling effort are greatest for the highest ability type. Thus, upon observing such a deviation to a higher signal, i.e., $s^P + \epsilon$, the incumbent needs to infer that the newcomer has talent c_L and react appropriately by choosing low effort (recall that now $c_I > \mu_N$). But then the talented newcomer has a strict incentive to actually deviate to $s^P + \epsilon$ for ϵ small enough, and thus no pooling equilibrium is possible. Consequently, if $c_I > \mu_N$, the game has a unique PBE consistent with D1 in which all newcomer types are perfectly revealed.

Denote the incumbent's beliefs after observing signaling effort s and when $c_I > \mu_N$ by $\rho(s|c_I > \mu_N) = \Pr[c_N = c_L|(s, c_I > \mu_N)]$. The incumbent's beliefs in the unique equilibrium are then

$$\rho(s|c_I > \mu_N) = \begin{cases} 1 & \text{if } s \geq \bar{s} \\ 0 & \text{else,} \end{cases}$$

and signaling efforts are $s_H = 0$ and $s_L = \bar{s}$. To the contrary, when $c_I \leq \mu_N$, only a pooling equilibrium exists in which

$$\rho(s|c_I \leq \mu_N) = \begin{cases} q & \text{if } s = 0 \\ \kappa \geq q & \text{else,} \end{cases}$$

and signaling efforts are $s_L = s_H = 0$.

Note that during the proof of the proposition we often used equilibrium dominance. When some message s' is equilibrium dominated only for one type of the newcomer and not for the other, the latter has clearly greater incentives to choose s'. Thus, equilibrium dominance in these cases suffices to show which equilibria are consistent with D1. These equilibria are then also consistent with the intuitive criterion (Cho and Kreps 1987), which relies on equilibrium dominance, implying D1 and the intuitive criterion select the same type of equilibrium. However, while inefficient pooling with $s^P > 0$ is not consistent with D1, it is with the intuitive criterion, because when $s^P > 0$ a deviation to $s = 0$ is not equilibrium dominated for either type of newcomer. Thus, we chose D1 as equilibrium refinement to further narrow down the set of reasonable equilibria.

Appendix D. Proof of Proposition 2: Monotone PBE

We first prove the impossibility of non-monotone equilibria. Then we continue and prove the properties of a PBE satisfying D1.

A. *Impossibility of Non-monotone PBE*

To prove that no PBE can be non-monotonic, i.e., also without using D1 as a refinement, we assume there are three types of newcomers with talent $0 < c_L < c_M < c_H$. This suffices to prove the result. A non-monotonic PBE would imply L and H pool, while M separates. Hence, there must be two levels of signaling, one for the type that separates, $s_S \geq 0$, and one for the two that pool, $s_P \geq 0$, where $s_P \neq s_S$.

After observing s_k, the incumbent chooses effort x_k. This implies in a non-monotonic equilibrium, the different types expected utilities are

$$u_L^{NM} = \left(1 - \sqrt{c_L x_P}\right)^2 - c_L s_P,$$

$$u_M^{NM} = \left(1 - \sqrt{c_M x_S}\right)^2 - c_M s_S,$$

$$u_H^{NM} = \left(1 - \sqrt{c_H x_P}\right)^2 - c_H s_P.$$

Payoffs from deviating from s_k to s_{-k} are

$$u_L^{dev} = \left(1 - \sqrt{c_L x_S}\right)^2 - c_L s_S,$$

$$u_M^{dev} = \left(1 - \sqrt{c_M x_P}\right)^2 - c_M s_P,$$

$$u_H^{dev} = \left(1 - \sqrt{c_H x_S}\right)^2 - c_H s_S.$$

For the non-monotonic PBE to exist, the following IC constraints need to hold:

$$u_L^{NM} \geq u_L^{dev} \Leftrightarrow \frac{\left(1 - \sqrt{c_L x_S}\right)^2 - \left(1 - \sqrt{c_L x_P}\right)^2}{c_L} \geq s_S - s_P,$$

$$u_M^{NM} \geq u_M^{dev} \Leftrightarrow \frac{\left(1 - \sqrt{c_M x_S}\right)^2 - \left(1 - \sqrt{c_M x_P}\right)^2}{c_M} \leq s_S - s_P,$$

$$u_H^{NM} \geq u_H^{dev} \Leftrightarrow \frac{\left(1 - \sqrt{c_H x_S}\right)^2 - \left(1 - \sqrt{c_H x_P}\right)^2}{c_H} \geq s_S - s_P.$$

Note that $K := \left(\left(1 - \sqrt{c_i x_S}\right)^2 - \left(1 - \sqrt{c_i x_P}\right)^2\right)/c_i$ is monotone in c_i, as

$$\frac{\partial K}{\partial c_i} = \frac{\sqrt{c_i}\left(\sqrt{x_S} - \sqrt{x_P}\right)}{c_i^2}.$$

Thus, when $u_L^{NM} \geq u_L^{dev}$ and $u_M^{NM} \geq u_M^{dev}$, it is impossible that $u_H^{NM} \geq u_H^{dev}$, and hence the IC constraints cannot simultaneously hold. This remains true for any distribution with at least three different types, and thus proves the result.

B. *Continuous Monotone PBE under D1*

We know from Lemma 2 that separation is only possible among types with talent greater than the incumbent, i.e., when $c_N < c_I$. Thus, because $c_L < c_I < c_H$ by Assumption 1, the newcomers with the lowest talent, $c_N \geq c_I$, need to pool, while others may reveal their types. Denote by $I_i^m\left(\underline{c}_i, \bar{c}_i\right)$, $\bar{c}_i \geq \underline{c}_i$, the ith interval on $\left[c_L, c_H\right]$, with 1 denoting the interval of least talented types that includes c_H. In these intervals the different types of newcomers choose signaling behavior that leads to $m \in \left\{P(\text{ooling}), S(\text{eparation})\right\}$, which implies they have to send a signal s_i^P if $c_N \in I_i^P\left(\underline{c}_i, \bar{c}_i\right)$ and a signal $s_i^S\left(c_N\right)$ if $c_N \in I_i^S\left(\underline{c}_i, \bar{c}_i\right)$.

We next show that it is not possible that there exists an interval of newcomer types that pool if there exists another interval of less able newcomers that separate in equilibrium. Assume to the contrary that there are I_i^S and I_{i+1}^P in equilibrium. Because separation is only possible among newcomers that are more able than the incumbent, this means that $\underline{c}_{i+1} < \bar{c}_{i+1} < c_I$; all newcomers in interval I_{i+1}^P are more talented than I. Denote the incumbent's effort when he believes to face $\kappa \in \left\{c_N, \left[\underline{c}_i, \bar{c}_i\right]\right\}$ after observing the newcomer's signaling by $x_I\left(\kappa\right)$ (that is the incumbent knows to either face c_N or newcomers distributed over some interval of types). Then $x_I\left(c_N\right)$, $c_N \leq \underline{c}_{i+1}$, is strictly smaller than $x_I\left(\left[\underline{c}_{i+1}, \bar{c}_{i+1}\right]\right)$, which follows from

$x_I(c_N)$ is strictly increasing in c_N for all $c_N < c_I$ (see Lemma 1) and the following lemma.

LEMMA 4: *In any pooling subgame equilibrium,*

$$\min\left\{\frac{\bar{c}_i}{(\bar{c}_i + c_I)^2}, \frac{\underline{c}_i}{(\underline{c}_i + c_I)^2}\right\} < x_I^{AI}\big([\underline{c}_i, \bar{c}_i]\big) < \frac{1}{4c_I}.$$

The effort in the pooling subgame equilibrium cannot be greater than the greatest effort expended under complete information against the same types, and it cannot be lower than the lowest effort expended under complete information against the same types.

PROOF:

We first prove the upper boundary. From Jensen's inequality it follows that

$$\sqrt{E\big[c_N|\bar{c}_i \geq c_N \geq \underline{c}_i\big]} > E\big[\sqrt{c_N}|\bar{c}_i \geq c_N \geq \underline{c}_i\big].$$

We can then replace the latter by the former in the formula determining A's strategy and get

$$\frac{E\big[c_N|\bar{c}_i \geq c_N \geq \underline{c}_i\big]}{\big(c_I + E\big[c_N|\bar{c}_i \geq c_N \geq \underline{c}_i\big]\big)^2} > x_I^{AI}\big([\underline{c}_i, \bar{c}_i]\big).$$

This expression is now greater than $x_I^{AI}\big([\underline{c}_i, \bar{c}_i]\big)$ and maximized when $E\big[c_N|\bar{c}_i \geq c_N \geq \underline{c}_i\big] = c_I$. In this case

$$\frac{c_I}{(c_I + c_I)^2} = \frac{1}{4c_I} > x_I^{AI}\big([\underline{c}_i, \bar{c}_i]\big),$$

proving the first half.

To prove the second half look at the first-order condition of the incumbent and then plug in the best response of the newcomer:

$$E\left[\sqrt{\frac{c_N}{x_I}} - c_I - c_N \middle| \bar{c}_i \geq c_N \geq \underline{c}_i\right] = 0.$$

Assume without loss of generality that $\bar{c}_i/(\bar{c}_i + c_I)^2 < \underline{c}_i/(\underline{c}_i + c_I)^2 \Leftrightarrow c_I^2/\bar{c}_i < \underline{c}_i$ and let $x_I = \bar{c}_i/(c_I + \bar{c}_i)^2$, that is the smallest effort chosen under complete information. Then we get, after some simplifications,

$$\text{(D1)} \qquad E\left[\left(\frac{\sqrt{c_N}(c_I + \bar{c}_i)}{\sqrt{\bar{c}_i}} - c_I - c_N\right)\middle| \bar{c}_i \geq c_N \geq \underline{c}_i\right]$$

$$=: E\big[\zeta(c_N, \bar{c}_i, c_I)\big|\bar{c}_i \geq c_N \geq \underline{c}_i\big] = 0.$$

Note that $\zeta(c_N, \bar{c}_i, c_I)$ has two roots in c_N, when $c_N = \bar{c}_i$ and when $c_N = c_I^2/\bar{c}_i$, and that

$$\left.\frac{\partial \zeta(c_N, \bar{c}_i, c_I)}{\partial c_N}\right|_{c_N=\bar{c}_i} < 0,$$

implying that as c_N gets smaller, ζ increases at this point. Consequently, for (D1) to be zero, we need that some types $c_N < c_I^2/\bar{c}_i$ occur with positive probability, but this contradicts $c_I^2/\bar{c}_i < \underline{c}_i$. Thus, evaluated at $x_I = \bar{c}_i/(c_I + \bar{c}_i)^2$, the incumbent's FOC is positive. Together with concavity of his utility function, this implies that in equilibrium $x_I^{AI}([\underline{c}_i, \bar{c}_i]) > \bar{c}_i/(c_I + \bar{c}_i)^2$, and thus it proves the second part of the lemma. ∎

Thus, indeed $x_I(c_N) < x_I([\underline{c}_{i+1}, \bar{c}_{i+1}])$ if $c_N \leq \underline{c}_{i+1}$. If $\underline{c}_{i+1} = c_L$ there exist unsent signals $s^u > s_{i+1}^P$. If $\underline{c}_{i+1} < c_L$, there must be another interval I_{i+2}^S of newcomers that separate. Since $x_I(\bar{c}_{i+2})$ strictly smaller than $x_I([\underline{c}_{i+1}, \bar{c}_{i+1}])$, incentive compatibility requires that the signal necessary to signal type $\bar{c}_{i+2}$, $s_{i+2}^S(\bar{c}_{i+2})$, must be strictly larger than s_{i+1}^P. Therefore, also in this case there exists a set of unsent signals $s^u \in (s_{i+1}^P, s_{i+2}^S(\bar{c}_{i+2}))$.

Returning to our analysis, Lemma 3 implies that from all $c_N \in I_{i+1}^P$, $\underline{c}_{i+1}$ has the greatest incentives to deviate to some $s' > s_{i+1}^P$. Because in both cases there exist unsent messages in this region, off equilibrium beliefs need to determine incentives. Whenever the incumbent observes some off equilibrium signaling effort $s' = s_{i+1}^P + \epsilon$, D1 requires that he needs to believe to face the type with the greatest incentives to send the signal, which is $\underline{c}_{i+1}$ (or potentially one of the even more talented newcomers, which would not change the conclusion). This gives $\underline{c}_{i+1}$ (and other talented newcomers in I_{i+1}^P) a strict incentive to deviate and thus the described situation cannot be a PBE satisfying D1. Consequently, we can rule out that there exist I_i^S and I_{i+1}^P in equilibrium. This leaves us with the following possible equilibria: all newcomers separate, all newcomers pool, and some relatively untalented newcomers pool, while the relatively talented newcomers separate, which proves the proposition. Note that our analysis implies that if there is a pooling interval with some threshold type $\underline{c} > c_L$ such that all $c_N < \underline{c}$ signal and reveal their types, the signaling effort cannot jump discretely at $\underline{c}$ for the same reason: when there are unsent signals, the most able among the pooling newcomers have an incentive to deviate to some unsent signal below the signal that separates the first type. Thus, under D1 any PBE must be continuous in the newcomer's type.

Appendix E. Proof of Theorem 1: Showing Off

Lemma 2 implies that only partial separation is possible whenever $c_I < c_H$. Thus, only types below some threshold c_Z may be able to signal their types in equilibrium. Recall that in Proposition 2 we have established that any PBE satisfying D1 must be a threshold equilibrium in which individuals' signaling messages are a continuous function of the newcomer's type. This in turn implies that to be indifferent between

signaling and abstaining from signaling, the effort of the incumbent when she knows she faces c_Z or when she does not know which newcomer she faces must be identical, $x_I^{AI}(c_Z) = x_I^{CI}(c_Z)$.

To study incentives we need to determine equilibrium effort under incomplete information. From Lemma 1 it follows that the incumbent, when he only knows that $c_N \in [c_Z, c_H]$, chooses the asymmetric information effort in the pooling equilibrium:

$$x_I^{AI}(c_Z) = \frac{\left(\int_{c_Z}^{c_H} \sqrt{c_N}\, dF(c_N)\right)^2}{\left(c_I\left(1 - F(c_Z)\right) + \int_{c_Z}^{c_H} c_N dF(c_N)\right)^2} = \frac{\left(E\left[\sqrt{c_N} | c_N \geq c_Z\right]\right)^2}{\left(c_I + E\left[c_N | c_N \geq c_Z\right]\right)^2}.$$

Recall that by Lemma 4 we have that

$$\min\left\{\frac{c_H}{(c_H + c_I)^2}, \frac{c_Z}{(c_Z + c_I)^2}\right\} < x_I^{AI}(c_Z) < \frac{1}{4c_I}.$$

The next lemma is crucial to prove existence and uniqueness of equilibrium.

LEMMA 5: *If $c_I \geq \mu_N = \sqrt{c_L c_H}$, $x_I^{AI}(c_Z)$ and $x_I^{CI}(c_Z)$ intersect exactly twice on $c_Z \in [c_L, c_H]$, once at some $c_Z = \tilde{c}_N \in \left(c_I^2/c_H, c_I\right)$, and once at $c_Z = c_H$. If $c_I < \mu_N$, there could be both one or two intersections, depending on the details of the distribution $F(c_N)$.*

PROOF:

Define

$$\Delta x_I(c_Z) \equiv x_I^{AI}(c_Z) - x_I^{CI}(c_Z). \tag{E1}$$

First note that

$$\lim_{c_Z \to c_H} \Delta x_I(c_Z) = 0,$$

which can be shown easily by repeatedly applying L'Hôpital's rule to $x_I^{AI}(c_Z)$. Hence, this is the first intersection of $x_I^{AI}(c_Z)$ and $x_I^{CI}(c_Z)$. Second, note that

$$\Delta x_I(c_I) = x_I^{AI}(c_I) - \frac{c_I}{(c_I + c_I)^2} = x_I^{AI}(c_I) - \frac{1}{4c_I} < 0$$

by Lemma 4. Moreover,

$$\Delta x_I\left(\frac{c_I^2}{c_H}\right) = x_I^{AI}\left(\frac{c_I^2}{c_H}\right) - \frac{c_I^2}{c_H\left(c_I + \frac{c_I^2}{c_H}\right)^2} = x_I^{AI}\left(\frac{c_I^2}{c_H}\right) - \frac{c_H}{(c_I + c_H)^2} > 0.$$

Hence, by continuity, there must be $\tilde{c}_N \in \left(c_I^2/c_H, c_I\right)$ such that $\Delta x_I(\tilde{c}_N) = 0$, proving existence of the second intersection for $c_I > \mu_N$. Note that when $c_I^2/c_H < c_L \Leftrightarrow c_I < \sqrt{c_L c_H} = \mu_N$, $\tilde{c}_N$ might be smaller than c_L, which is why there might be only one intersection on $[c_L, c_H]$ in this case.

Could there be another $c_Z \in (\tilde{c}_N, c_H)$ such that $x_I^{AI} = x_I^{CI}$? Next we show that this is not possible. First note that x_I^{CI} is strictly concave, increasing on $[c_L, c_I)$ and decreasing on $(c_I, c_H]$. To see how x_I^{AI} changes with c_Z, we differentiate it:

$$\frac{\partial x_I^{AI}}{\partial c_Z} = \frac{2f(c_Z)\left(\int_{c_Z}^{c_H}\sqrt{c_N}f(c_N)\,dc_N\right)}{\left(c_I\int_{c_Z}^{c_H}f(c_N)\,dc_N + \int_{c_Z}^{c_H}c_N f(c_N)\,dc_N\right)^3}$$

$$\times\left[(c_I + c_Z)\int_{c_Z}^{c_H}\sqrt{c_N}f(c_N)\,dc_N\right.$$

$$\left.-\sqrt{c_Z}\left(\int_{c_Z}^{c_H}c_N f(c_N)\,dc_N + c_I\big(1 - F(c_Z)\big)\right)\right].$$

We can easily verify that the sign of the derivative turns on the sign of the squared brackets, which turns on $x_I^{AI} - x_I^{CI}$. Thus,

$$\operatorname{sign}\left[\frac{\partial x_I^{AI}}{\partial c_Z}\right] = \operatorname{sign}\left[x_I^{AI} - x_I^{CI}\right],$$

which implies that at $\tilde{c}_N$ the derivative is zero. Since x_I^{CI} is increasing on $[c_L, c_I]$, x_I^{AI} must be decreasing on some interval $(\tilde{c}_N, c_Z']$, $c_Z' > c_I$, and it can only increase again after $x_I^{AI} = x_I^{CI}$ once more. If this were too happen for some $c_Z' < c_H$, x_I^{AI} were increasing on $(c_Z', c_H]$, because x_I^{CI} is still strictly decreasing. But this would imply that when $c_Z = c_H$, $x_I^{AI} > x_I^{CI}$, which cannot be the case. Hence, c_Z' cannot be smaller than c_H, implying x_I^{AI} decreases on $(\tilde{c}_N, c_H)$, and there cannot be $c_Z \in (\tilde{c}_N, c_H)$ such that $x_I^{AI} = x_I^{CI}$.

Finally, note that it must be that $x_I^{AI} > x_I^{CI}$ if $c_Z \in (c_Z', \tilde{c}_N)$ for some $c_Z' < \tilde{c}_N$. Could it be the case that there exists $c_Z' > c_L$ such that both efforts intersect again? Note that x_I^{CI} increases on $[c_L, \tilde{c}_Z]$ and is concave. This implies that if there is another intersection, x_I^{AI} must be steeper than x_I^{CI} at this point and it thus must have a strictly positive slope. However, such an intersection then implies that x_I^{AI} cannot be differentiable at the intersection, because we know that the derivative must be zero when $x_I^{AI} = x_I^{CI}$. However, it is differentiable. Thus, this intersection cannot exist. Consequently, $x_I^{AI}(c_Z) = x_I^{CI}(c_Z)$ on $[c_L, c_H]$ exactly twice when $c_I \geq \mu_N$ and once or twice else. ∎

Note that the preceding discussion together with Lemma 3 implies that also in the general case there cannot be equilibria in which pooling newcomers choose a costly signal, $s^P > 0$. By Lemma 3 we know that the least able newcomer has the greatest incentive to deviate to $s = 0$. Because $\tilde{c}_N > c_I^2/c_H \Leftrightarrow c_I < \sqrt{\tilde{c}_N c_H}$ by Lemma 5, the incumbent will choose the lowest effort under complete information against the least able newcomer with talent c_H for all $c_N \in [\tilde{c}_N, c_H]$. Thus, if she sees $s = 0$ off the equilibrium path, she will infer to face the least able newcomer, to which she responds by choosing low efforts. This gives all pooling newcomers an

incentive to deviate, and thus there cannot be an equilibrium in which the pooling newcomers engange in signaling.

To sum up, the threshold type needs to be indifferent between signaling and pooling. By the arguments above we know that the threshold type $\tilde{c}_N$ incurs no signaling costs. Since he only reacts optimally to the incumbent's effort, the incumbent needs to choose identical effort against the threshold type whether he signals his type or pools, $\Delta x_I(\tilde{c}_N) = 0$. From Lemma 2 it follows that the threshold must be lower than c_I. From Lemma 5 it follows that if $c_I \geq \mu_N$, a unique threshold type exists for which this is the case, while for $c_I < \mu_N$ there is either a unique interior threshold type or no newcomer chooses a costly signal in equilibrium.

Equilibrium Signaling.—To find the signal cost function $s(c')$ for which truth-telling is strictly optimal, we take the first-order condition of $\pi_N^S(c'; c_N)$ in (B1) with respect to the stated type c':

$$-\frac{2\left(c_I - \sqrt{c_N c'} + c'\right)^2}{\left(c_I + c'\right)^3} + \frac{2\left(1 - \frac{c_N}{2\sqrt{c_N c'}}\right)\left(c_I - \sqrt{c_N c'} + c'\right)}{\left(c_I + c'\right)^2} - c_N s'(c').$$

We are searching for a function for which utility is maximized when $c' = c_N$. Thus, now let $c' = c_N$ and set the derivative equal to zero to find the local extrema:

$$\frac{c_I(c_N - c_I)}{(c_I + c_N)^3} - c_N s'(c_N) = 0.$$

This is a differential equation that can easily be solved. The general solution for $s(c')$ is

$$\tilde{s}(c') = K - \frac{2c_I + c'}{(c_I + c')^2} + \frac{\ln\left(\frac{c_I + c'}{c'}\right)}{c_I},$$

where K is a constant. As discussed earlier, $\tilde{c}_N$ will not expend signaling effort in equilibrium and thus $s(\tilde{c}_N) = 0$. This implies

$$K = \frac{2c_I + \tilde{c}_N}{(c_I + \tilde{c}_N)^2} - \frac{\ln\left(\frac{c_I + \tilde{c}_N}{\tilde{c}_N}\right)}{c_I},$$

and therefore equilibrium signaling follows

$$s^*(c') = \frac{\ln\left(\frac{\tilde{c}_N(c_I + c')}{c'(c_I + \tilde{c}_N)}\right)}{c_I} + \frac{2c_I + \tilde{c}_N}{(c_I + \tilde{c}_N)^2} - \frac{2c_I + c'}{(c_I + c')^2}.$$

It remains to verify that $s^*(c')$ really implements a separating equilibrium. Take (B1) and replace $s(c')$ by $s^*(c')$:

$$\pi_N^*(c';c_N) = \frac{\left(c_I - \sqrt{c_N c'} + c'\right)^2}{\left(c_I + c'\right)^2} - c_N\left(\frac{\ln\left(\frac{\tilde{c}_N(c_I+c')}{c'(c_I+\tilde{c}_N)}\right)}{c_I} + \frac{2c_I + \tilde{c}_N}{\left(c_I + \tilde{c}_N\right)^2} - \frac{2c_I + c'}{\left(c_I + c'\right)^2}\right).$$

Maximizing this with respect to the claimed type c', we get the first-order condition

$$\frac{\partial \pi^*(c';c_N)}{\partial c'} = -\frac{c_N^2(c' - c_I)\left(\sqrt{c_N c'} - c'\right)}{\left(c_I + c'\right)^2 \left(c_N c'\right)^{3/2}} = 0.$$

This equation has two roots, one at $c' = c_I$ and one at $c' = c_N$. The second-order conditions are

$$\left.\frac{\partial^2 \pi^*(c';c_N)}{\partial c'^2}\right|_{c'=c_I} = \frac{c_N\left(c_I - \sqrt{c_I c_N}\right)}{4\sqrt{c_I^7 c_N}} > 0 \quad \forall c_N \in \left[c_L, \tilde{c}_N\right]$$

and

$$\left.\frac{\partial^2 \pi^*(c';c_N)}{\partial c'^2}\right|_{c'=c_N} = \frac{c_N - c_I}{2c_N\left(c_I + c_N\right)^2} < 0 \quad \forall c_N \in \left[c_L, \tilde{c}_N\right],$$

respectively. Thus, the latter, where $c' = c_N$, is indeed a local maximum as $\tilde{c}_N < c_I$, while $c' = c_I$ is a local minimum. Moreover, since there are no other roots of the first-order condition, we can also exclude corner-solutions, because the utility minimum occurs when $c' = c_I$, which is not within the range of potential types, implying $\pi_N^*(c';c_N)$ is strictly quasi-concave in c' on $\left[c_L, \tilde{c}_N\right]$. Could a talented newcomer benefit from deviating by pretending to have $c_N > \tilde{c}_N$? No, because this would give him the exact same utility he would get from pretending to have type $\tilde{c}_N$ by definition of this type, because the incumbent's effort and the cost from signaling would be just the same. Hence, given $s^*(c')$, truthfully revealing one's type is the unique best strategy for all types $\left[c_L, \tilde{c}_N\right]$.

Incentive Compatibility for Pooling Newcomers.—Next we look at incentive compatibility for the pooling newcomer types $\left[\tilde{c}_N, c_H\right]$. We need to assure that none of them has an incentive to mimic any $c_N < \tilde{c}_N$. The benefit from mimicking is that I competes less fiercely against the more efficient types than in the pooling equilibrium, which is beneficial. Consider the incentives of some $c_N > \tilde{c}_N$ to mimic some $c_N' < \tilde{c}_N$. The utility he gets from pooling is

$$\pi_N^{Pool} = \frac{\sqrt{\frac{x_I^{AI}}{c_N}} - x_I^{AI}}{\sqrt{\frac{x_I^{AI}}{c_N}} - x_I^{AI} + x_I^{AI}} - c_N\left(\sqrt{\frac{x_I^{AI}}{c_N}} - x_I^{AI}\right) = \left(1 - \sqrt{c_N x_I^{AI}}\right)^2.$$

In Lemma 3 we have already established that more able newcomers have the greatest incentive to deviate from pooling to some greater signal. Thus, the threshold type is the newcomer type with the greatest incentives to deviate. Since $\tilde{c}_N$ has no incentive to mimic a given $c_N' < \tilde{c}_N$, neither has another type with $c_N > \tilde{c}_N$. Thus, we established existence of equilibrium with N-threshold newcomer type $\tilde{c}_N$.

Next consider comparative statics. Condition (E1) determines the N-threshold. Using the implicit function theorem, we know that

$$\left.\frac{dc_Z}{dc_I}\right|_{c_Z=\tilde{c}_N} = -\left.\frac{\frac{\partial \Delta x_I}{\partial c_I}}{\frac{\partial \Delta x_I}{\partial c_Z}}\right|_{c_Z=\tilde{c}_N}$$

or

$$\left.\frac{dc_Z}{dc_I}\right|_{c_Z=\tilde{c}_N} = -\frac{\dfrac{2\left(1-F(\tilde{c}_N)\right)\left(\int_{\tilde{c}_N}^{c_H}\sqrt{c}\,dF(c)\right)^2}{\left(c_I\left(1-F(\tilde{c}_N)\right)+\int_{\tilde{c}_N}^{c_H}c\,dF(c)\right)^3} - \dfrac{2\tilde{c}_N}{(c_I+\tilde{c}_N)^3}}{\dfrac{2f(\tilde{c}_N)\left(\int_{\tilde{c}_N}^{c_H}\sqrt{c}\,dF(c)\right)\left(c_I\sqrt{\tilde{c}_N}\left(1-F(\tilde{c}_N)\right)-(c_I+\tilde{c}_N)\int_{\tilde{c}_N}^{c_H}\sqrt{c}\,dF(c)+\sqrt{\tilde{c}_N}\int_{\tilde{c}_N}^{c_H}c\,dF(c)\right)}{\left(c_I\left(1-F(\tilde{c}_N)\right)+\int_{\tilde{c}_N}^{c_H}c\,dF(c)\right)^3} + \dfrac{c_I-\tilde{c}_N}{(c_I+\tilde{c}_N)^3}}.$$

First note that the denominator is strictly positive since $c_I > \tilde{c}_N$ and $c_I\sqrt{\tilde{c}_N}\left(1-F(\tilde{c}_N)\right) - (c_I+\tilde{c}_N)\int_{\tilde{c}_N}^{c_H}\sqrt{c}\,dF(c) + \sqrt{\tilde{c}_N}\int_{\tilde{c}_N}^{c_H}c\,dF(c) = 0$. Thus, the sign of the derivative turns on the numerator's sign. The numerator is positive if and only if

$$\frac{2\left(1-F(\tilde{c}_N)\right)\left(\int_{\tilde{c}_N}^{c_H}\sqrt{c}\,dF(c)\right)^2}{\left(c_I\left(1-F(\tilde{c}_N)\right)+\int_{\tilde{c}_N}^{c_H}c\,dF(c)\right)^3} - \frac{2\tilde{c}_N}{(c_I+\tilde{c}_N)^3} > 0$$

$$\Leftrightarrow \frac{\left(1-F(\tilde{c}_N)\right)\left(\int_{\tilde{c}_N}^{c_H}\sqrt{c}\,dF(c)\right)^2}{\left(c_I\left(1-F(\tilde{c}_N)\right)+\int_{\tilde{c}_N}^{c_H}c\,dF(c)\right)^3} - \frac{\tilde{c}_N}{(c_I+\tilde{c}_N)^3} > 0$$

$$\Leftrightarrow \frac{\left(1-F(\tilde{c}_N)\right)}{\left(c_I\left(1-F(\tilde{c}_N)\right)+\int_{\tilde{c}_N}^{c_H}c\,dF(c)\right)}x_I^{AI} - x_I^{CI}\frac{1}{c_I+\tilde{c}_N} > 0$$

$$\Leftrightarrow \frac{\left(1-F(\tilde{c}_N)\right)}{\left(c_I\left(1-F(\tilde{c}_N)\right)+\int_{\tilde{c}_N}^{c_H}c\,dF(c)\right)} > \frac{1}{c_I+\tilde{c}_N}$$

$$\Leftrightarrow \frac{1}{\left(c_I+\frac{\int_{\tilde{c}_N}^{c_H}c\,dF(c)}{\left(1-F(\tilde{c}_N)\right)}\right)} > \frac{1}{c_I+\tilde{c}_N} \Leftrightarrow \frac{\int_{\tilde{c}_N}^{c_H}c\,dF(c)}{\left(1-F(\tilde{c}_N)\right)} < \tilde{c}_N.$$

But this is impossible. Thus, the numerator must be negative, and therefore the derivative is positive: $\tilde{c}_N$ increases in c_I.

For low enough c_I, the threshold is lower than c_L and thus no signaling takes place. Denote the first c_I such that no signaling takes place by $\tilde{c}_I$. This $\tilde{c}_I$ solves

$$\frac{c_L}{(c_L+\tilde{c}_I)^2} = \frac{\left(E\left[\sqrt{c_N}\right]\right)^2}{\left(\tilde{c}_I+E\left[c_N\right]\right)^2} \Leftrightarrow \tilde{c}_I = \frac{\sqrt{c_L}E\left[c_N\right]-c_L E\left[\sqrt{c_N}\right]}{E\left[\sqrt{c_N}\right]-\sqrt{c_L}} > 0.$$

Note, $\tilde{c}_I$ is the generalization of μ_N from the binary case. Hence, if $c_I > \tilde{c}_I$, the most able newcomer types signal, while otherwise all newcomer types abstain from signaling and no information is transmitted.

Appendix F. Proof of Proposition 3: The Incumbent's Perspective—Binary Case

Using equilibrium utilities given in Appendix B, the difference in the incumbent's expected utility between complete and asymmetric information is equal to

$$\begin{aligned}\Delta\pi_I = \pi_I^{CI}-\pi_I^{AI} = {} & \frac{q(1-q)\left(\sqrt{c_H}-\sqrt{c_L}\right)\left(c_I-\sqrt{c_H c_L}\right)}{(c_I+c_H)^2(c_I+c_L)^2\left(c_I+(1-q)c_H+qc_L\right)^2} \\ & \times\Big[c_I^3\left(2\sqrt{c_H c_L}+c_L+c_H\right) \\ & \quad + c_I^2\Big(\left(c_H+c_L+(1-q)c_H+qc_L\right)\sqrt{c_H c_L}+4c_H c_L\Big) \\ & \quad + \Big(c_I^3+2qc_I^2+c_H c_L\left(3c_I-\sqrt{c_H c_L}\right)\Big)\left((1-q)c_H+qc_L\right)\Big].\end{aligned}$$

If $c_I = \sqrt{c_H c_L}$, $\Delta\pi_I = 0$ and the incumbent is indifferent between the newcomer signaling or laying low. For $c_I > \sqrt{c_H c_L}$, she prefers to receive information about the newcomer's type. Finally, for c_I smaller than $\sqrt{c_H c_L}$, the sign of the difference also depends on $3c_I - \sqrt{c_H c_L}$, which is decreasing in c_H and thus smallest for the highest possible c_H. Using $c_H = 2c_L$, which is the largest admissible value of c_H, it becomes $3c_I - \sqrt{2}\,c_L > 0$ by Assumption 1. Thus, for $c_I < \sqrt{c_H c_L}$, $\Delta\pi_I < 0$ and the incumbent is better off when the newcomer lays low.

Appendix G. Proof of Proposition 4: The Incumbent's Perspective—General Case

Recall that by Lemma 1,

$$E\pi_I = \int_{c_L}^{c_Z}\frac{c_N^2}{(c_N+c_I)^2}dF(c_N) + \frac{\left(\int_{c_Z}^{c_H}\sqrt{c_N}\,dF(c_N)\right)^2\int_{c_Z}^{c_H}c_N\,dF(c_N)}{\left(c_I\left(1-F(c_Z)\right)+\int_{c_Z}^{c_H}c_N\,dF(c_N)\right)^2}.$$

Differentiating with respect to threshold c_Z yields

$$\text{(G1)} \quad \frac{\partial E\pi_I}{\partial c_Z}$$

$$= f(c_Z)\left(\frac{c_Z^2}{(c_I+c_Z)^2} - \frac{2\sqrt{c_Z}\gamma_2(c_Z)\gamma_1(c_Z)}{\left(c_I+\gamma_1(c_Z)\right)^2} + \frac{\gamma_2(c_Z)^2\left[(2c_I+c_Z)\gamma_1(c_Z) - c_I c_Z\right]}{\left(c_I+\gamma_1(c_Z)\right)^3}\right),$$

where $\gamma_1(c_Z) \equiv \int_{c_Z}^{c_H} c_N dF(c_N)/\left(1-F(c_Z)\right)$ and $\gamma_2(c_Z) \equiv \int_{c_Z}^{c_H}\sqrt{c_N}\,dF(c_N)/\left(1-F(c_Z)\right)$. For the rest of the proof we omit the argument of γ_1 and γ_2, c_Z, for better readability of the equations whenever possible. Using $x_I^{CI}(c_I,c_Z) = c_Z/(c_I+c_Z)^2$ and $x_I^{AI}(c_I,c_Z) = (\gamma_2)^2/(c_I+\gamma_1)^2$, we can write the FOC for an extremum as

$$\frac{\partial E\pi_I}{\partial c_Z}$$

$$= f(c_Z)\left(c_Z x_I^{CI}(c_I,c_Z) - \frac{2\sqrt{c_Z}\gamma_1}{\gamma_2}x_I^{AI}(c_I,c_Z) + \frac{\left((2c_I+c_Z)\gamma_1 - c_I c_Z\right)}{(c_I+\gamma_1)}x_I^{AI}(c_I,c_Z)\right)$$

$$= f(c_Z)c_Z\left(\sqrt{x_I^{CI}(c_I,c_Z)} - \sqrt{x_I^{AI}(c_I,c_Z)}\right)$$

$$\times\left(\sqrt{x_I^{CI}(c_I,c_Z)} + \sqrt{x_I^{AI}(c_I,c_Z)} - \frac{2\gamma_1}{\gamma_2\sqrt{c_Z}}\frac{x_I^{AI}(c_I,c_Z)}{\sqrt{x_I^{CI}(c_I,c_Z)}}\right) = 0,$$

and thus we have a local extremum when $x_I^{CI}(c_I,c_Z) = x_I^{AI}(c_I,c_Z)$. By Lemma 5 this happens only at the N-threshold and when $c_Z = c_H$. Next we show that this is the only possible type of extremum in the relevant range, i.e., we show that

$$\zeta(c_Z) \equiv f(c_Z)\,c_Z\left(\sqrt{x_I^{CI}(c_I,c_Z)} + \sqrt{x_I^{AI}(c_I,c_Z)} - \frac{2\gamma_1}{\gamma_2\sqrt{c_Z}}\frac{x_I^{AI}(c_I,c_Z)}{\sqrt{x_I^{CI}(c_I,c_Z)}}\right) < 0$$

$$\Leftrightarrow \sqrt{x_I^{CI}(c_I,c_Z)} + \sqrt{x_I^{AI}(c_I,c_Z)} - \frac{2\gamma_1}{\gamma_2\sqrt{c_Z}}\frac{x_I^{AI}(c_I,c_Z)}{\sqrt{x_I^{CI}(c_I,c_Z)}} < 0$$

for $c_L \leq c_Z < c_H$. First note that this expression is negative when $x_I^{CI}(c_I,c_Z) = x_I^{AI}(c_I,c_Z)$ because

$$\text{(G2)} \qquad \frac{2\gamma_1}{\gamma_2\sqrt{c_Z}} > 2$$

as $\gamma_2 > \sqrt{c_Z}$ and $\gamma_1 > (\gamma_2)^2$ by Jensen's inequality. More generally the expression is always negative when $x_I^{CI} \leq x_I^{AI}$, which is true for any $c_Z < \tilde{c}_Z$ by Lemma 5. Thus, the N-threshold constitutes the only extremum on $c_Z \in [c_L, \tilde{c}_Z]$. To prove that it is also negative for all $c_Z \geq \tilde{c}_Z$, we will construct bounds on the inequality

in (G2). First we multiply by $\sqrt{x_I^{CI}(c_I, c_Z)}$ and insert back the expressions for $x_I^{AI}(c_I, c_Z)$ and $x_I^{CI}(c_I, c_Z)$ to get

$$\frac{c_Z}{(c_I + c_Z)^2} + \frac{\gamma_2}{c_I + \gamma_1}\frac{\sqrt{c_Z}}{c_I + c_Z} - \frac{2\gamma_1}{\sqrt{c_Z}}\frac{\gamma_2}{(c_I + \gamma_1)^2} < 0.$$

Now we multiply by $(c_I + \gamma_1)^2$ to get

$$\frac{(c_I + \gamma_1)^2}{(c_I + c_Z)^2}c_Z + \frac{(c_I + \gamma_1)}{c_I + c_Z}\gamma_2\sqrt{c_Z} - \frac{2\gamma_1}{\sqrt{c_Z}}\gamma_2 < 0.$$

Note that $(c_I + \gamma_1)/(c_I + c_Z)$ is decreasing in c_I, and thus, in order to bound the left-hand side of the inequality (and show that it cannot be positive), we can insert the lower bound $c_I = c_L$ in this expression. This lower bound is smallest for a given c_H when c_L is the smallest possible (according to Assumption 1), and thus we use $c_I = c_L = c_H/2$:

$$\frac{\left(\frac{c_H}{2} + \gamma_1\right)^2}{\left(\frac{c_H}{2} + c_Z\right)^2}c_Z + \frac{\left(\frac{c_H}{2} + \gamma_1\right)}{\frac{c_H}{2} + c_Z}\gamma_2\sqrt{c_Z} - \frac{2\gamma_1}{\sqrt{c_Z}}\gamma_2 < 0.$$

Furthermore, we can bound c_H in $((c_H/2) + \gamma_1)/((c_H/2) + c_Z)$ by γ_1 as $c_H \geq \gamma_1$, and thus our inequality, after simplifying and dividing by γ_1, becomes

$$\frac{\gamma_1\left(\frac{3}{2}\right)^2}{\left(\frac{\gamma_1}{2} + c_Z\right)^2}c_Z + \frac{\left(\frac{3}{2}\right)}{\frac{\gamma_1}{2} + c_Z}\gamma_2\sqrt{c_Z} - \frac{2}{\sqrt{c_Z}}\gamma_2 < 0.$$

Next note that the last two terms are linear and decreasing in γ_2, and thus we can bound the inequality further by setting $\gamma_2 = \sqrt{c_Z}$, its lower bound, to get

$$\frac{\gamma_1\left(\frac{3}{2}\right)^2}{\left(\frac{\gamma_1}{2} + c_Z\right)^2}c_Z + \frac{\left(\frac{3}{2}\right)}{\frac{\gamma_1}{2} + c_Z}c_Z - 2 < 0.$$

Finally, the first two terms are decreasing in γ_1 and thus we can bound the inequality by setting $\gamma_1 = c_Z$, its lower bound, to get

$$\frac{c_Z\left(\frac{3}{2}\right)^2}{\left(\frac{3}{2}c_Z\right)^2}c_Z + \frac{\left(\frac{3}{2}\right)}{\frac{3}{2}c_Z}c_Z - 2 = 0.$$

Thus, except in the limit case, in which $c_Z = c_H$, the inequality in (G2) is strict. As a consequence, all extrema of the incumbent's expected utility function must fulfill $x_I^{CI} = x_I^{AI}$. From Lemma 5 we know that this happens no more than twice: at the N-threshold and at $c_Z = c_H$.

In order to check the SOC we need to check the second derivative of the incumbent's expected utility with respect to c_Z at the N-threshold. This reveals that

$$\begin{aligned}\left.\frac{\partial^2 E\pi_I}{\partial c_Z^2}\right|_{c_Z=\tilde{c}_Z} &= \zeta'(c_Z)\left(\left.\sqrt{x_I^{CI}}\right|_{c_Z=\tilde{c}_Z} - \left.\sqrt{x_I^{AI}}\right|_{c_Z=\tilde{c}_Z}\right) \\ &\quad + \zeta(c_Z)\left(\left.\frac{\partial\sqrt{x_I^{CI}}}{\partial c_Z}\right|_{c_Z=\tilde{c}_Z} - \left.\frac{\partial\sqrt{x_I^{AI}}}{\partial c_Z}\right|_{c_Z=\tilde{c}_Z}\right) \\ &= \zeta(c_Z)\left.\frac{\partial\sqrt{x_I^{CI}}}{\partial c_Z}\right|_{c_Z=\tilde{c}_Z}.\end{aligned}$$

Because $\frac{\partial\sqrt{x_I^{CI}(c_I,c_Z)}}{\partial c_Z} = \frac{1}{2\sqrt{x_I^{CI}(c_I,c_Z)}}\frac{\partial x_I^{CI}(c_I,c_Z)}{\partial c_Z} > 0$ and

$$\frac{\partial\sqrt{x_I^{AI}(c_I,c_Z)}}{\partial c_Z} = \frac{1}{2\sqrt{x_I^{AI}(c_I,c_Z)}}\frac{\partial x_I^{AI}(c_I,c_Z)}{\partial c_Z} = 0$$

at the N-threshold (see Lemma 5) and $\zeta(c_Z) < 0$, $(\partial^2 E\pi_I/\partial c_Z^2)|_{c_Z=\tilde{c}_Z} < 0$, and thus the N-threshold constitutes a global expected utility maximum for the incumbent. Note that we have shown that the N-threshold is a local maximum also when $4c_L \geq c_H > c_L$, which is more permissive than Assumption 1. However, in this case we cannot exclude that the incumbent prefers another threshold type $c_Z > \tilde{c}_Z$ over the N-threshold.

Appendix H. Proof of Theorem 2: Norms and Aggregate Effort

First we show how norms, which burden a newcomer that chooses positive signaling effort with a fixed cost $\xi \geq 0$, impact the threshold type.

PROPOSITION 7: *Assume signaling is sanctioned by a fixed cost $\xi \geq 0$. Equilibrium remains a threshold equilibrium and the fraction of newcomers that choose positive signaling effort and thus separate in equilibrium strictly decreases in ξ until no newcomers separate anymore. More precisely, increasing the fixed cost strictly increases the talent of the threshold type until it reaches c_L, and no newcomer separates anymore.*

PROOF:

First note that our above analysis showing that equilibrium must be a threshold equilibrium with some threshold type c_Z remains valid also when there are fixed costs to signaling. We thus only need to show how the threshold type changes

with ξ. The threshold type c_Z is implicitly determined by the following indifference condition:

$$\left(1-\sqrt{c_Z x_I^{CI}(c_Z)}\right)^2-\xi-\left(1-\sqrt{c_Z x_I^{CI}(c_Z)}\right)^2 = 0.$$

To see how c_Z changes, we use the implicit function theorem. Taking into account that when $\xi = 0$ we have $c_Z = \tilde{c}_N$ and thus also $x_I^{CI}(c_Z) = x_I^{AI}(c_Z)$, and thus by Lemma 5 also $\partial x_I^{AI}(c_Z)/\partial c_Z = 0$, we get

$$\left.\frac{\partial c_Z}{\partial \xi}\right|_{\xi=0} = \frac{x_I^{CI}(c_Z)}{\sqrt{c_N x_I^{CI}(c_Z)}\left(\sqrt{c_N x_I^{CI}(c_Z)}-1\right)\frac{c_I-c_Z}{(c_I+c_Z)^2}}.$$

In an interior equilibrium, $1/c_N > x_I^{CI}(c_Z)$ implies $\sqrt{c_N x_I^{CI}(c_Z)}-1<0$, and thus $(\partial c_Z/\partial\xi)|_{\xi=0} < 0$, because also $c_I > c_Z = \tilde{c}_N$. Thus, the threshold type must decrease (become more talented) when ξ increases from zero. Moreover, note that the threshold comparative statics must be monotonic: higher fixed costs lead to a more able threshold type. Assume to the contrary they were non-monotonic. Then there exists some type c_N' who is the threshold type for (at least) two different levels of ξ. But the threshold type uniquely determines equilibrium spending by the incumbent, and thus the contest utility of c_N' when signaling and when laying low must be the same in both situations. Then he cannot be indifferent between signaling and pooling for different levels of fixed costs, implying the same type cannot be a threshold type twice. Thus, the threshold must be monotonically increasing until it reaches c_L and no newcomer separates anymore. ∎

Now recall that because of Lemma 1, expected aggregate effort given a threshold type c_Z is

$$EX = \int_{c_L}^{c_Z}\frac{1}{c_N+c_I}dF(c_N)+\frac{\int_{c_Z}^{c_H}\sqrt{c_N}\,dF(c_N)\int_{c_Z}^{c_H}\frac{1}{\sqrt{c_N}}dF(c_N)}{\left(c_I\left(1-F(c_Z)\right)+\int_{c_Z}^{c_H}c_N\,dF(c_N)\right)}.$$

Taking the derivative with respect to c_Z yields

$$\frac{\partial EX}{\partial c_Z}$$

$$= f(c_Z)\left(\frac{(c_I+c_Z)\,\gamma_2(c_Z)\,\gamma_3(c_Z)}{\left(\gamma_1(c_Z)+c_I\right)^2}-\frac{\sqrt{c_Z}\,\gamma_3(c_Z)}{\left(\gamma_1(c_Z)+c_I\right)}-\frac{\gamma_2(c_Z)}{\sqrt{c_Z}\left(\gamma_1(c_Z)+c_I\right)}+\frac{1}{c_I+c_Z}\right),$$

where $\gamma_3(c_Z) \equiv \left(\int_{c_Z}^{c_H}f(c_N)/\sqrt{c_N}\,dc_N\right)/\left(1-F(c_Z)\right)$ and, as before, $\gamma_1(c_Z) \equiv \int_{c_Z}^{c_H}c_N dF(c_N)/\left(1-F(c_Z)\right)$ and $\gamma_2(c_Z) \equiv \int_{c_Z}^{c_H}\sqrt{c_N}\,dF(c_N)/\left(1-F(c_Z)\right)$. As before, for better readability we suppress the argument of γ_1, γ_2, and γ_3 whenever possible. Rewriting this derivative in terms of incumbent effort against the threshold

newcomer $x_I^{CI}(c_I, c_Z)$ and incumbent effort under asymmetric information given threshold c_Z, $x_I^{AI}(c_I, c_Z)$, we get

$$\frac{\partial EX}{\partial c_Z} = f(c_Z)\left(\sqrt{x_I^{CI}(c_I, c_Z)} - \sqrt{x_I^{AI}(c_I, c_Z)}\right)\left(\frac{1}{\sqrt{c_Z}} - \frac{(c_I + c_Z)\gamma_3}{(\gamma_1 + c_I)}\right),$$

and thus we again have a local extremum when $x_I^{CI}(c_I, c_Z) = x_I^{AI}(c_I, c_Z)$. Again, by Lemma 5 this happens only at the N-threshold and when $c_Z = c_H$. Next we show that this is the only possible type of extremum in the relevant range, i.e., we show that

$$\frac{1}{\sqrt{c_Z}} \neq \frac{(c_I + c_Z)\gamma_3}{(\gamma_1 + c_I)}.$$

Rewriting, the sign of the expression in brackets turns on

$$\frac{(\gamma_1 + c_I)}{(c_Z + c_I)} - \sqrt{c_Z}\gamma_3.$$

Note that $\sqrt{c_Z}\gamma_3 = \int_{c_Z}^{c_H}\left(\sqrt{c_Z}/\sqrt{c_N}\right)f(c_N)dc_N/\left(1 - F(c_Z)\right) < 1$ when $c_Z < c_H$. Furthermore, $\gamma_1 > c_Z$. Thus, the first term is larger than one and the second one smaller than one, and thus it always holds that

$$\frac{(\gamma_1 + c_I)}{(c_Z + c_I)} - \sqrt{c_Z}\gamma_3 > 0.$$

Next take a look at the SOC at the extrema, i.e., when $x_I^{CI}(c_I, c_Z) = x_I^{AI}(c_I, c_Z)$. The second derivative of expected aggregate effort with respect to c_Z simplifies considerably to

$$\left.\frac{\partial^2 EX}{\partial c_Z^2}\right|_{x_I^{CI} = x_I^{AI}} = f(c_Z)\frac{\partial\left(\sqrt{x_I^{CI}(c_I, c_Z)}\right)}{\partial c_Z}\left(\frac{1}{\sqrt{c_Z}} - \frac{(c_I + c_Z)\gamma_3}{(\gamma_1 + c_I)}\right),$$

where we used $\partial x_I^{AI}(c_Z)/\partial c_Z = 0$ when $x_I^{CI}(c_I, c_Z) = x_I^{AI}(c_I, c_Z)$ (see Lemma 5) as well as the FOC. Note that $\partial\sqrt{x_I^{CI}(c_I, c_Z)}/\partial c_Z > 0$ at the N-threshold. Thus, $\partial^2 EX/\partial c_Z^2|_{c_Z = \tilde{c}_N} > 0$ and the N-threshold constitutes a global expected aggregate effort minimum.

Appendix I. Proof of Theorem 3: Norms and Selection

First note that we established in Proposition 7 in Appendix H that stricter norms decrease the threshold newcomer type. Next we consider the effect of a decrease of this threshold type on expected efficiency.

We define expected efficiency as the expected marginal cost of effort of the contest winner given threshold newcomer type c_Z, i.e.:

$$
\begin{aligned}
EE(c_Z) &\equiv E\big[c_i | i \text{ wins}\big] \\
&= \int_{c_L}^{c_H} \left(\frac{x_N(c_Z, c_N)}{x_N(c_Z, c_N) + x_I(c_Z, c_N)} c_N + \frac{x_I(c_Z, c_N)}{x_N(c_Z, c_N) + x_I(c_Z, c_N)} c_I \right) dF(c_N) \\
&= \int_{c_L}^{c_H} \left(\frac{x_N(c_Z, c_N)}{x_N(c_Z, c_N) + x_I(c_Z, c_N)} c_N \right. \\
&\qquad \left. + \left(1 - \frac{x_I(c_Z, c_N)}{x_N(c_Z, c_N) + x_I(c_Z, c_N)} \right) c_I \right) dF(c_N) \\
&= \int_{c_L}^{c_H} \frac{x_N(c_Z, c_N)}{x_N(c_Z, c_N) + x_I(c_Z, c_N)} (c_N - c_I)\, dF(c_N) + c_I \\
&= \int_{c_L}^{c_Z} (c_N - c_I) \frac{x_N^{CI}(c_N)}{x_N^{CI}(c_N) + x_I^{CI}(c_N)}\, dF(c_N) \\
&\quad + \int_{c_Z}^{c_H} (c_N - c_I) \frac{x_N^{AI}(c_N)}{x_N^{AI}(c_N) + x_I^{AI}(c_Z)} dF(c_N) + c_I \\
&= \int_{c_L}^{c_Z} (c_N - c_I) \frac{c_I}{c_I + c_N} dF(c_N) \\
&\quad + \int_{c_Z}^{c_H} (c_N - c_I) \left(1 - \sqrt{c_N} \frac{\gamma_2(c_Z)}{c_I + \gamma_1(c_Z)} \right) dF(c_N) + \ c_I,
\end{aligned}
$$

where $\gamma_1(c_Z)$ and $\gamma_2(c_Z)$ are defined as before. Using

$$
\begin{aligned}
\frac{\partial \frac{\gamma_2(c_Z)}{c_I + \gamma_1(c_Z)}}{\partial c_Z} &= -f(c_Z) \frac{\big(c_I + \gamma_1(c_Z)\big)\sqrt{c_Z} - (c_I + c_Z)\gamma_2(c_Z)}{\big(1 - F(c_Z)\big)\big(c_I + \gamma_1(c_Z)\big)^2} \\
&= -\frac{f(c_Z)(c_I + c_Z)}{\big(1 - F(c_Z)\big)\big(c_I + \gamma_1(c_Z)\big)} \left(\frac{\sqrt{c_Z}}{(c_I + c_Z)} - \frac{\gamma_2(c_Z)}{\big(c_I + \gamma_1(c_Z)\big)} \right) \\
&= \frac{f(c_Z)(c_I + c_Z)}{\big(1 - F(c_Z)\big)\big(c_I + \gamma_1(c_Z)\big)} \left(\sqrt{x_I^{AI}(c_Z)} - \sqrt{x_I^{CI}(c_N)} \right),
\end{aligned}
$$

the derivative of $EE(c_Z)$ with respect to c_Z, can be written as

$$\begin{aligned}
\frac{\partial EE(c_Z)}{\partial c_Z} &= f(c_Z)\Bigg[(c_Z - c_I)\frac{c_I}{c_I + c_Z} - (c_Z - c_I)\Big(1 - \sqrt{c_Z}\,\frac{\gamma_2}{c_I + \gamma_1}\Big) \\
&\qquad - \int_{c_Z}^{c_H}(c_N - c_I)\sqrt{c_N}\,\frac{\partial \frac{\gamma_2(c_Z)}{c_I + \gamma_1(c_Z)}}{\partial c_Z}\,dF(c_N)\,\frac{1}{f(c_Z)}\Bigg] \\
&= f(c_Z)\Bigg[(c_Z - c_I)\sqrt{c_Z}\Big(\sqrt{x_I^{AI}(c_Z)} - \sqrt{x_I^{CI}(c_N)}\Big) \\
&\qquad - \int_{c_Z}^{c_H}(c_N - c_I)\sqrt{c_N}\,\frac{f(c_Z)(c_I + c_Z)}{\big(1 - F(c_Z)\big)\big(c_I + \gamma_1(c_Z)\big)} \\
&\qquad\qquad \times \Big(\sqrt{x_I^{AI}(c_Z)} - \sqrt{x_I^{CI}(c_N)}\Big)\,dF(c_N)\,\frac{1}{f(c_Z)}\Bigg] \\
&= f(c_Z)\Big(\sqrt{x_I^{AI}(c_Z)} - \sqrt{x_I^{CI}(c_N)}\Big) \\
&\quad \times \Bigg((c_Z - c_I)\sqrt{c_Z} \\
&\qquad - \int_{c_Z}^{c_H}(c_N - c_I)\sqrt{c_N}\,\frac{(c_I + c_Z)}{\big(1 - F(c_Z)\big)\big(c_I + \gamma_1(c_Z)\big)}\,dF(c_N)\Bigg).
\end{aligned}$$

We know from Lemma 5 that except at the N-threshold and at $c_Z = c_H$, $x_I^{AI}(c_Z) \neq x_I^{CI}(c_Z)$, and thus the N-threshold constitutes a local extremum of $EE(c_Z)$. Any other extremum in the relevant region must satisfy

$$\text{(I1)} \qquad (c_Z - c_I)\sqrt{c_Z} - \frac{(c_I + c_Z)}{\big(c_I + \gamma_1(c_Z)\big)}\,\frac{\int_{c_Z}^{c_H}(c_N - c_I)\sqrt{c_N}\,dF(c_N)}{\big(1 - F(c_Z)\big)} = 0.$$

At $c_Z = c_I < c_H$, this cannot be the case, because

$$-\frac{2c_I}{\big(c_I + \gamma_1(c_I)\big)}\,\frac{\int_{c_I}^{c_H}(c_N - c_I)\sqrt{c_N}\,dF(c_N)}{\big(1 - F(c_I)\big)} < 0.$$

If we can show that the left-hand side of equation (I1) is strictly negative for all $c_Z < c_I$, we prove that the N-threshold is the only extremum on $[c_L, c_I]$.

First note that $(c_Z - c_I)\sqrt{c_Z}$ is negative and strictly increasing in c_Z when $c_Z < c_I \leq 2c_Z$ which is the relevant range taking into account Assumption 1. Also note that the second term is $\big((c_I + c_Z)/\big(c_I + \gamma_1(c_Z)\big)\big)E\big[(c_N - c_I)\sqrt{c_N}|c_N \geq c_Z\big]$, where $(c_I + c_Z)/\big(c_I + \gamma_1(c_Z)\big) \leq 1$ and strictly so for $c_Z < c_H$. We know already that for $c_Z = c_I$ the left-hand side of equation (I1) is negative. Now decrease c_Z

starting at this point. Then the left term turns and remains strictly negative, while the right term might be positive, zero, or negative, depending on c_I and c_Z. As long as the right term is positive or zero, the difference must remain negative, because we subtract a positive term from a negative term. If the second term turns negative as well for some c_Z, we must compare their magnitudes. Because $(c_Z - c_I)\sqrt{c_Z}$ is negative and strictly increases in c_Z in the relevant range it holds that

$$\begin{aligned} 0 &> \frac{(c_I + c_Z)}{\left(c_I + \gamma_1(c_Z)\right)} E\left[(c_N - c_I)\sqrt{c_N} \,|\, c_N \geq c_Z\right] \\ &> E\left[(c_N - c_I)\sqrt{c_N} \,|\, c_N \geq c_Z\right] > \sqrt{c_Z}(c_Z - c_I), \end{aligned}$$

and thus the difference remains strictly negative (as long as $c_Z < c_H$). Consequently, there is no other extremum for $c_Z < c_I$ apart from the N-threshold. Using $x_I^{AI}(c_Z) = x_I^{CI}(c_Z)$ as well as $\partial x_I^{AI}(c_Z)/\partial c_Z = 0$ when $c_Z = \tilde{c}_N$ (see Lemma 5),

$$\begin{aligned} \left.\frac{\partial^2 EE(c_Z)}{\partial c_Z^2}\right|_{c_Z=\tilde{c}_N} &= -f(c_Z)\frac{\partial\left(\sqrt{x_I^{CI}(c_I, c_Z)}\right)}{\partial c_Z} \\ &\quad \times \left((c_Z - c_I)\sqrt{c_Z} - \frac{c_I + c_Z}{c_I + \gamma_1(c_Z)} \frac{\int_{c_Z}^{c_H}(c_N - c_I)\sqrt{c_N}\, dF(c_N)}{\left(1 - F(c_Z)\right)}\right). \end{aligned}$$

We know from our above analysis that $\partial\left(\sqrt{x_I^{CI}(c_I, c_Z)}\right)/\partial c_Z > 0$, and the expression in parentheses is negative on $c_Z \in [c_L, c_I]$, and therefore $\partial^2 EE(c_Z)/\partial c_Z^2|_{c_Z=\tilde{c}_N} > 0$, proving that the N-threshold maximizes expected efficiency (minimizes the expected marginal cost of the winner) in that range.

Appendix J. Proof of Proposition 5: Overconfidence

To prove the result regarding overconfidence, consider the following modification of the model. As before, the newcomer can be of two types. He is talented with probability q or untalented with probability $1 - q$. He is always fully aware of his true ability when he is talented, but he may exhibit overconfidence when untalented. In particular, with probability $b \leq 1 - q$ he is untalented but wrongly believes to be talented and does not consider the possibility of being overconfident. As such an untalented and overconfident newcomer, he will choose an effort level such that he incurs the same effort cost a talented type would incur in equilibrium. The idea behind this is that he sees himself working as hard as the talented type, but the resulting effective effort x_i will be only $x_i(c_L/c_H) < x_i$. Thus, while the overconfident newcomer thinks he is productive, he is not. We assume that ex ante the incumbent holds biased beliefs as well and thus thinks she is facing a talented newcomer with probability $q + b$ and an untalented one with probability $1 - q - b$.

The newcomer's incentives to signal his type are unchanged. Whether signaling happens turns only on the relation of the incumbent's talent c_I to the geometric mean of the newcomer's potential types $\mu_N = \sqrt{c_H c_L}$ and not the respective probabilities with which these occur. Thus, a newcomer that believes to be talented signals his type if and only if $c_I > \mu_N$, which follows from Proposition 1. In this case, the overconfident newcomer types flop and learn through signaling that they are in fact untalented. A truly able newcomer, to the contrary, reveals his high talent.

If $c_I > \mu_N$, the newcomer's flopping, or not, will be observed by the incumbent as well. Given the equilibrium beliefs derived in the proof of Proposition 1, after observing a lower level of signaling the incumbent is convinced to face a low ability newcomer. Thus, after signaling, both competitors choose efforts under full information. If $c_I \leq \mu_N$, the newcomer will not signal his type in equilibrium and thus no information is revealed. In particular, the incumbent still believes that the newcomer is talented with probability $q + b$ while a fraction b of newcomer types holds the incorrect belief that they are talented even though they are in fact not.

We now prove Proposition 5 for $b = 1 - q$. By continuity our results are also valid in a neighborhood of $b = 1 - q$ for b sufficiently large. Under signaling full information prevails and we can use our previous results to derive equilibrium efforts. Expected aggregate effort equals

$$EX^{CI} = \frac{c_I + qc_H + (1-q)c_L}{(c_I + c_H)(c_I + c_L)}.$$

When no signaling takes place, the incumbent wrongly believes the likelihood of a talented opponent is $q + b$ and thus maximizes

$$\max_{x_I} (q+b)\frac{x_I}{x_I + x_L} + (1 - q - b)\frac{x_I}{x_I + x_H} - c_I x_I.$$

An untalented, non-overconfident newcomer chooses effort to

$$\max_{x_H} \frac{x_H}{x_I + x_H} - c_H x_H,$$

a talented one to

$$\max_{x_L} \frac{x_L}{x_I + x_L} - c_L x_L,$$

while an untalented, overconfident newcomer chooses

$$x_{HL} = \frac{c_L}{c_H} x_L.$$

This yields the following equilibrium efforts:

$$x_I^{AI} = \frac{c_L}{(c_I + c_L)^2}, \quad x_L^{AI} = \frac{c_I}{(c_I + c_L)^2}, \quad x_{HL}^{AI} = \frac{c_L}{c_H}\frac{c_I}{(c_I + c_L)^2}.$$

Expected aggregate effort in this case is

$$EX^{AI} = \frac{(c_I + c_H)\, c_L + q c_I (c_H + c_L)}{c_H (c_I + c_L)^2}.$$

Thus, the difference in aggregate effort, after simplifying, is

$$\Delta EX = \frac{(c_H - c_L)(c_I^2 - c_H c_L)(1 - q)}{c_H (c_I + c_H)(c_I + c_L)^2},$$

which turns only on the sign of $c_I^2 - c_H c_L$. In particular, aggregate effort is higher for $c_I > \sqrt{c_H c_L}$ under complete information and vice versa if no information is transmitted, which is in contrast to the case with $b = 0$. By continuity there exists at least one switching point for each c_I such that going from $b = 0$ to $b = 1 - q$ reverses the inequality.

When the incumbent does not learn the newcomer's type through signaling, she overestimates the likelihood of facing a strong newcomer. This tends to dampen her effort if the newcomer is exceptional and increases her effort if the newcomer is nonexceptional. For a large enough share of overconfident newcomers, the incumbent's effort is thus higher under signaling if and only if the newcomer is exceptional. Moreover, the overconfident newcomer chooses a different effort level than if he was aware of his true strength. His effort level will be higher than when he knows he is untalented if and only if the incumbent is quite talented ($c_I < \mu_N$). Thus, only in this case will the newcomer be more aggressive if his bias is not revealed which is exactly what happens in equilibrium. The reason is that in this case a talented newcomer competes more fiercely with the incumbent than an untalented one (as they are more similar), and thus the overconfident newcomer expends more than if he knew he was untalented. Thus, in the case when there is no signaling and thus no learning of types, i.e., $c_I < \mu_N$, the biased newcomer overinvests in effort to the benefit of the principal. So far we have assumed that also the incumbent holds biased beliefs. This is not necessary for Proposition 5 to hold, though. The same effects can be generated when the incumbent correctly anticipates the newcomer's behavior and best-responds to it. In this case, the newcomer chooses the efforts characterized above, unaware of his bias. The incumbent on the other hand, best-responds to the newcomer's choices. Equilibrium expressions of efforts are available, but the expressions are extremely unwieldy. Thus, we show that our results are robust to this setting by example. In particular, consider the case where $c_L = 1$, $c_H = 2$, and $q = 0.5$. Figure 5 shows the difference between aggregate effort under complete information and absent information transmission for $b = 0$, $b = 1/10$, and $b = 3/10$ as a function of c_I. For $b = 1/10$ the results are qualitatively unchanged, while for $b = 0.3$ results are exactly reversed.

Appendix K. Proof of Proposition 6: Productive Signal

To prove the result we proceed in two steps. First, we derive a bound on the effectiveness κ of the productive signal that guarantees that no newcomer type

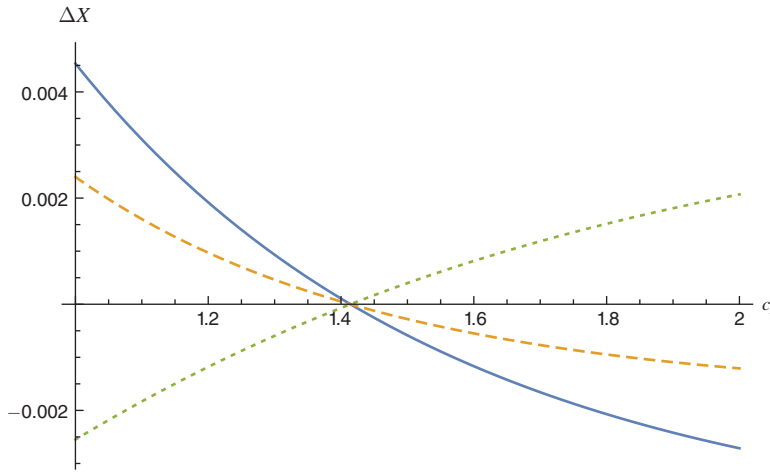

FIGURE 5

Notes: The blue line shows the difference in expected aggregate effort between full information and asymmetric information for the case where no newcomer is overconfident ($b = 0$). The orange (dashed) line shows this difference for a small fraction of overconfident newcomers ($b = 1/10$), while the green (dotted) line shows this difference for the case of a large fraction of overconfident newcomers ($b = 3/10$). As before, when $c_I = \sqrt{c_L c_H} = \mu_N = \sqrt{2}$ the differences are zero.

finds it profitable to use the productive signal to commit effort greater than the contest efforts in the equilibrium we have studied so far. We then prove that incentive compatibility still works and signaling works exactly as it did in the Baseline model.

Commitment.—Assume that signaling effort s_i translates into contest effort at rate $s_i \kappa$. Thus, the newcomer's total contest effort equals $s_N \kappa + x_N$. When $\kappa > 0$, the newcomer can effectively pre-commit to effort because he can move first. Note that among all efforts some type of newcomer could pre-commit to, the best is the effort he would choose as a Stackelberg leader (see Yildirim 2005). This effort is $x_N^{SB} = c_I/\left(4c_N^2\right)$. While generally pre-commitment of effort might lead to corner solutions, in which the player who moves second stays passive, Assumption 1 assures an interior equilibrium.

It is well known since Dixit (1987) that in a game like the one we are studying, a player benefits from commiting to an effort level that is greater than the level of effort he would choose in a standard Nash equilibrium, if the player is "stronger" than his opponent. Otherwise, he would like to commit to lower effort. The fact that the newcomer chooses effort also in the contest itself prevents him from under-commiting effort, because he would just add as much effort as needed to achieve his Nash equilibrium effort in the normal contest stage (see Lemma 2 of Yildirim 2005). However, the newcomer might overcommit effort, which would change the contest equilibrium. Therefore we need to find a condition under which this is not in the newcomer's interest.

As we have established earlier, the newcomer's Nash equilibrium effort is $x_N^{CI} = c_I/(c_I + c_N)^2$. If he overcommits relative to this effort using the signal s', $\kappa s' \geq c_I/(c_I + c_N)^2$, in the contest itself he would choose zero effort, while the incumbent simply best responds (see again Lemma 2 in Yildirim 2005). In this case, his utility from choosing the signal would be

$$\pi_N^s = \frac{\kappa s'}{\kappa s' + \sqrt{\frac{\kappa s'}{c_I}} - \kappa s'} - c_N s' = \sqrt{c_I \kappa s'} - c_N s'.$$

Taking the first derivative with respect to s' yields

$$\frac{\partial \pi_N^s}{\partial s'} = \frac{1}{2}(c_I \kappa s')^{-1/2}(c_I \kappa) - c_N.$$

Evaluating this at $s' = \frac{1}{\kappa} x_N^{CI} = \frac{1}{\kappa} \frac{c_I}{(c_I + c_N)^2}$ this derivative must not be positive. Thus, we need

$$\frac{1}{2}\left(c_I \kappa \frac{1}{\kappa} \frac{c_I}{(c_I + c_N)^2}\right)^{-1/2}(c_I \kappa) - c_N \leq 0 \Leftrightarrow$$

$$\kappa \leq \frac{2c_N}{c_I}\left(\frac{c_I^2}{(c_I + c_N)^2}\right)^{1/2} = \frac{2c_N}{(c_I + c_N)}.$$

Without loss of generality let $c_I = \lambda c_N$, $\lambda \in (1/2, 2)$ (by Assumption 1). Then,

$$\kappa \leq \frac{2c_N}{(c_I + c_N)} = \frac{2c_N}{(\lambda c_N + c_N)} = \frac{2}{(\lambda + 1)}.$$

Since this has to hold for all $\lambda \in (1/2, 2)$, a sufficient condition is $\kappa \leq \bar{\kappa} = 2/3$. Thus, when $\kappa \leq \bar{\kappa} = 2/3$, the newcomer never finds it profitable to precommit to an effort level that is higher than his complete information Nash equilibrium effort and the subsequent contest stage remains unchanged relative to the baseline model.

Incentive Compatibility.—Now let us study the signaling game. Assume the newcomer chooses signal s'. From Lemma 2 in Yildirim (2005) it follows that, if the newcomer doesn't overcommit effort and information is revealed truthfully, then the newcomer simply adds effort to reach the Nash equilibrium level, $x'_N = x_N^{CI} - \kappa s'$. Likewise, the incumbent chooses Nash equilibrium efforts. Thus, along the equilibrium path, the newcomer chooses efforts

$$x'_L(s_L) = \frac{c_I}{(c_I + c_L)^2} - \kappa s_L \quad \text{and} \quad x'_H(s_H) = \frac{c_I}{(c_I + c_H)^2} - \kappa s_H,$$

the incumbent chooses Nash equilibrium efforts, and thus utilities for the newcomer types, from the perspective of the signaling stage, are

$$\tilde{\pi}_L^{CI} = \frac{\kappa s_L + x_L'(s_L)}{\kappa s_L + x_L'(s_L) + x_I^{CI}(c_I, c_L)} - c_L x_L'(s_L) - c_L s_L$$

$$= \frac{c_I^2}{(c_I + c_L)^2} - c_L s_L(1 - \kappa),$$

$$\tilde{\pi}_H^{CI} = \frac{\kappa s_H + x_H'(s_H)}{\kappa s_H + x_H'(s_H) + x_I^{CI}(c_I, c_H)} - c_H x_H'(s_H) - c_H s_H$$

$$= \frac{c_I^2}{(c_I + c_H)^2} - c_H s_H(1 - \kappa).$$

Off equilibrium information is not truthfully revealed. Denote $x_I^{CI}(c_I, c_j)$ the incumbent's effort if he believes to face a newcomer of type j and denote $BR_i\big(x_I^{CI}(c_I, c_j); s_i\big)$ newcomer type i's best response against this effort of the incumbent, when he spent already s_i on a signal. Then,

$$BR_L\big(x_I^{CI}(c_I, c_H); s_H\big) = \sqrt{\frac{x_I^{CI}(c_I, c_H)}{c_L}} - x_I^{CI}(c_I, c_H) - \kappa s_H,$$

$$BR_H\big(x_I^{CI}(c_I, c_L); s_L\big) = \sqrt{\frac{x_I^{CI}(c_I, c_L)}{c_H}} - x_I^{CI}(c_I, c_L) - \kappa s_L.$$

This implies the newcomer's utility, again from the perspective of the signaling stage, is as follows:

$$\tilde{\pi}_L^D = \frac{BR_L\big(x_I^{CI}(c_I, c_H); s_H\big) + \kappa s_H}{BR_L\big(x_I^{CI}(c_I, c_H); s_H\big) + \kappa s_H + x_I^{CI}(c_I, c_H)} - c_L BR_L\big(x_I^{CI}(c_I, c_H); s_H\big) - c_L s_H$$

$$= \frac{\big(c_H + c_I - \sqrt{c_L c_H}\big)^2}{(c_H + c_I)^2} - c_L s_H(1 - \kappa),$$

$$\tilde{\pi}_H^D = \frac{BR_H\big(x_I^{CI}(c_I, c_L); s_L\big) + \kappa s_L}{BR_H\big(x_I^{CI}(c_I, c_L); s_L\big) + \kappa s_L + x_I^{CI}(c_I, c_L)} - c_H BR_H\big(x_I^{CI}(c_I, c_L); s_L\big) - c_H s_L$$

$$= \frac{\big(c_L + c_I - \sqrt{c_L c_H}\big)^2}{(c_L + c_I)^2} - c_H s_L(1 - \kappa).$$

In equilibrium it must be the case that $s_H = 0$. Then, incentive compatibility requires

$$\frac{c_I^2}{(c_I + c_L)^2} - c_L s_L(1-\kappa) \geq \frac{(c_H + c_I - \sqrt{c_L c_H})^2}{(c_H + c_I)^2}$$

$$\frac{c_I^2}{(c_I + c_H)^2} \geq \frac{(c_L + c_I - \sqrt{c_L c_H})^2}{(c_L + c_I)^2} - c_H s_L(1-\kappa).$$

Note that, if $\kappa = 0$, this condition is exactly the condition for separation we found above with binary types. Moreover, as it was the case there, there exists $s_L > 0$ such that separation is possible if and only if $c_I > \sqrt{c_L c_H}$. Finally, also note that when s_L is chosen such that the untalented newcomer's IC constraint is just binding,

$$s_L(\kappa) = \frac{\frac{(c_I - \sqrt{c_H c_L} + c_L)^2}{(c_I + c_L)^2} - \frac{c_I^2}{(c_I + c_H)^2}}{c_H(1-\kappa)},$$

the incumbent's IC constraint holds and $\kappa \cdot s_L(\kappa) < x_L^{CI}$, which can be checked readily. It is easily checked that $s_L(\kappa)$ increases in κ, implying signaling increases as it becomes more productive. If we restrict the incumbent's off equilibrium beliefs using D1 as before, the game has a unique equilibrium in which the talented newcomer signals his type through costly signaling and types separate. Similar steps establish that, again as before with $\kappa = 0$, a separating equilibrium cannot exist if $c_I < \sqrt{c_L c_H}$.

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